MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NOR, KEY CODE AME

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. This document establishes the requirements and test procedures for the acquisition of an electronic NOR gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0719 failures per million operating hours for class I or 0.3555 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be AME.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A .

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/249-1 for class I and M28787/249-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3 to 18 V ±5%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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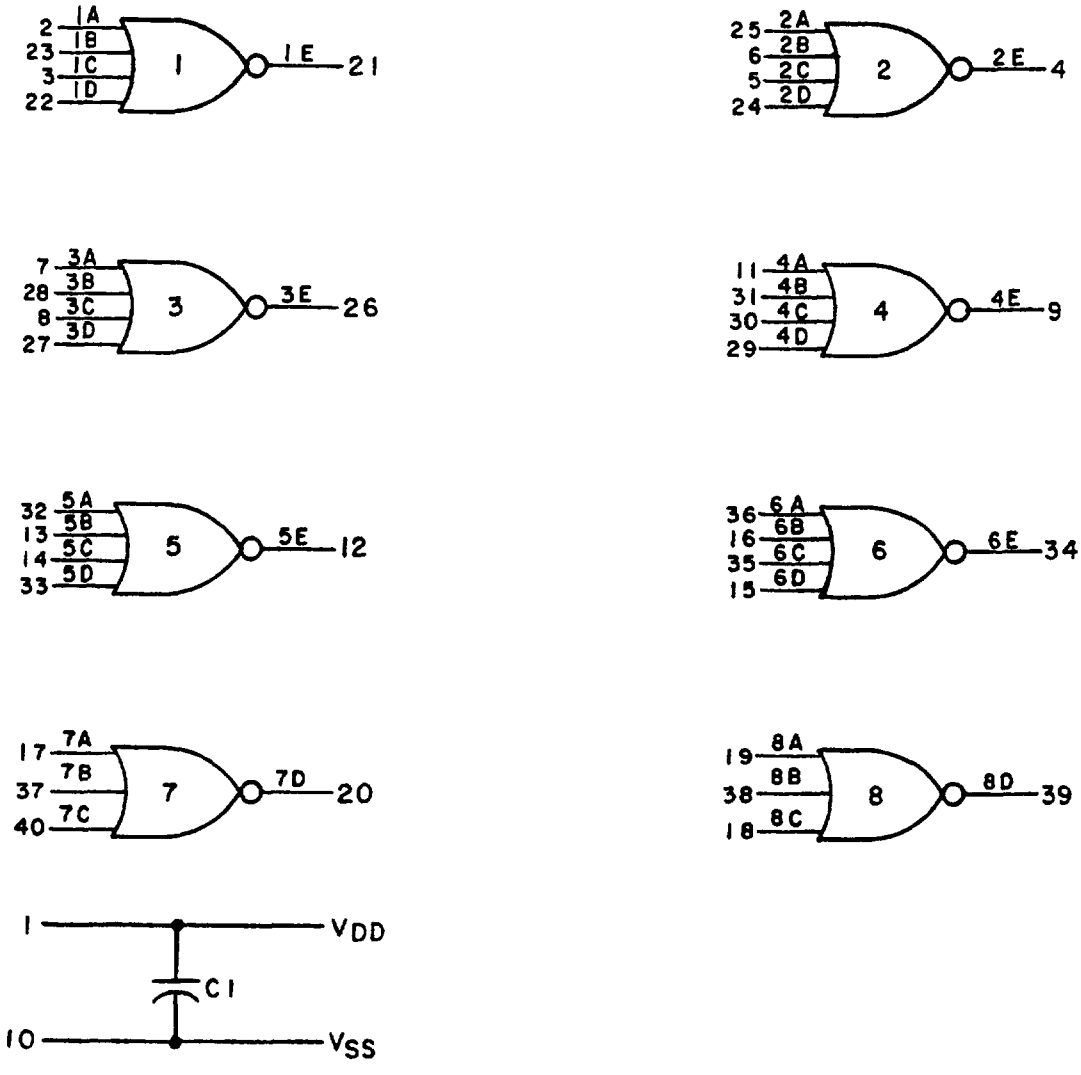


FIGURE 1. AME module function diagram, NOR gate.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B30	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C30	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D16	Functional	VI	← See table VI →											
E1-E8	Output voltage (low level), I_{OL} = 0	VII	X	0	50	0	50	0	50	0	50	0	50	mV
F1-F8	Output voltage (high level), I_{OH} = 0	VIII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
G1-G8	Output voltage (low level), V_{DD} = 5 V	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H8	Output voltage (high level), V_{DD} = 5 V	X	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
J1-J8	Output voltage (low level), V_{DD} = 15 V	XI	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
K1-K8	Output voltage (high level), V_{DD} = 15 V	XII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
L1-L30	Input current, high level	XIII	X	0	200	0	200	0	200	0	200	0	200	nA
M1-M30	Input current, low level	XIV	X	0	-200	0	-200	0	-200	0	-200	0	-200	nA
N1	Power supply current, low level	XV	X	0	8	0	8	0	80	0	8	0	130	µA
N2	Power supply current, high level	XV	X	0	8	0	8	0	80	0	8	0	130	µA
P1-P8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XVI	S	20	200	20	200	22	220	15	150	24	235	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P9-P16	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XVI	Q	68	680	68	680	75	750	51	515	80	812	ns
Q1-Q8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XVII	S	20	200	20	200	22	220	15	150	24	235	ns
Q9-Q16	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	68	680	68	680	75	750	51	515	80	802	ns
R1-R8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XVIII	Q	10	100	10	100	11	110	7	75	12	120	ns
S1-S8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XIX	Q	10	100	10	100	11	110	7	75	12	120	ns
T1-T8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XX	Q	8	80	8	80	9	90	6	60	10	96	ns
U1-U8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XXI	Q	8	80	8	80	9	90	6	60	10	96	ns
V1-V30	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXII	S	32	320	32	320	35	355	24	245	39	373	ns
V31-V60	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXII	Q	52	520	52	520	57	575	39	395	63	612	ns
W1-W30	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXIII	S	32	320	32	320	35	355	24	245	39	377	ns
W31-W60	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXIII	Q	52	520	52	520	57	575	39	395	63	612	ns
X1-X30	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXIV	Q	13	130	13	130	14	145	10	100	16	155	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1-Y30	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXV	Q	13	130	13	130	14	145	10	100	16	155	ns
Z1-Z30	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXVI	Q	10	100	10	100	11	110	7	75	12	120	ns
AA1-AA30	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXVII	Q	10	100	10	100	11	110	7	75	12	120	ns
BB	Power supply filter capacitance	XXVIII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
CC1-CC30	Input capacitance	XXIX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature test. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through C in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through K and N through CC shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2. Test conditions L and M shall be performed at 25°C, 60°C, and 85°C only, using the initial limits for the applicable operating temperature.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition BB, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through CC shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

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	Class I	Class II	Unit
Maximum power dissipation:	2.4	2.4	mW
Input voltage:			
Low level, V_{DD} = 5 V	0.0 min, 1.5 max	0.0 min, 1.5 Max	V
Low level, V_{DD} = 15 V	0 min, 4 max	0.0 min, 1.5 max	V
High level, V_{DD} = 5 V	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
High level, V_{DD} = 15 V	11 min, V_{DD} max	11 min, V_{DD} max	V
Input current:			
Low level (V_{IL} = 0 V) V_{DD} = 18 V	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18 V) V_{DD} = 18 V	0 min, 200 max	0 min, 200 max	nA
Output drive current (I_{OL}), V_{DD} = 5 V:			
I_{OL} = (V_{OL} = 0.4 V) at -55°C	---	0.64 min	mA
0°C and 25°C	0.51 min	0.51 min	mA
60°C	0.44 min	0.44 min	mA
85°C	---	0.36 min	mA
I_{OH} (V_{OH} = 4.6 V) at -55°C	---	-0.25 min	mA
0°C and 25°C	-0.20 min	-0.20 min	mA
60°C	-0.17 min	-0.17 min	mA
85°C	---	-0.14 min	mA

6.5 Module function. The module contains six independent 4 input CMOS NOR gates and two independent 3 input CMOS NOR gates.

6.5.1 Input to output relationship. The "E" outputs shall have the input to output relationship of table IIA for positive logic. The "D" outputs shall have the input to output relationship of table IIB for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs to obtain extended drive and sink capability is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

a. 1 and 2.
b. 3 and 4.
c. 5 and 6.
d. 7 and 8.

6.5.3 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.
b. Soldering-iron tips, metal parts of fixtures, and tools, and handling facilities should be grounded.
c. Signals should not be applied to the inputs while the device power supply (V_{DD}) is off.
d. All input signals and conditioning voltages must be removed before V_{DD} is disconnected.
e. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the input protection diodes.
f. All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

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6.5.4 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$P_{Total} = (CV_{DD}{}^2 f)N + \text{quiescent power dissipation.}$$

where C = load capacitance.
V_{DD} = supply voltage.
f = operating frequency.
N = number of circuits.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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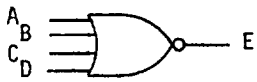
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TABLE IIA. Input to output relationship for "E" outputs. 1/, 2/

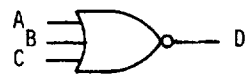
Inputs				Output
A	B	C	D	E
L	L	L	L	H
H	L	L	L	L
L	H	L	L	L
H	H	L	L	L
L	L	H	L	L
H	L	H	L	L
L	H	H	L	L
H	H	H	L	L
L	L	L	H	L
H	L	L	H	L
L	H	L	H	L
H	H	L	H	L
L	L	H	H	L
H	L	H	H	L
L	H	H	H	L
H	H	H	H	L



$E = \overline{A+B+C+D}$

TABLE IIB. Input to output relationship for "D" outputs. 1/, 2/

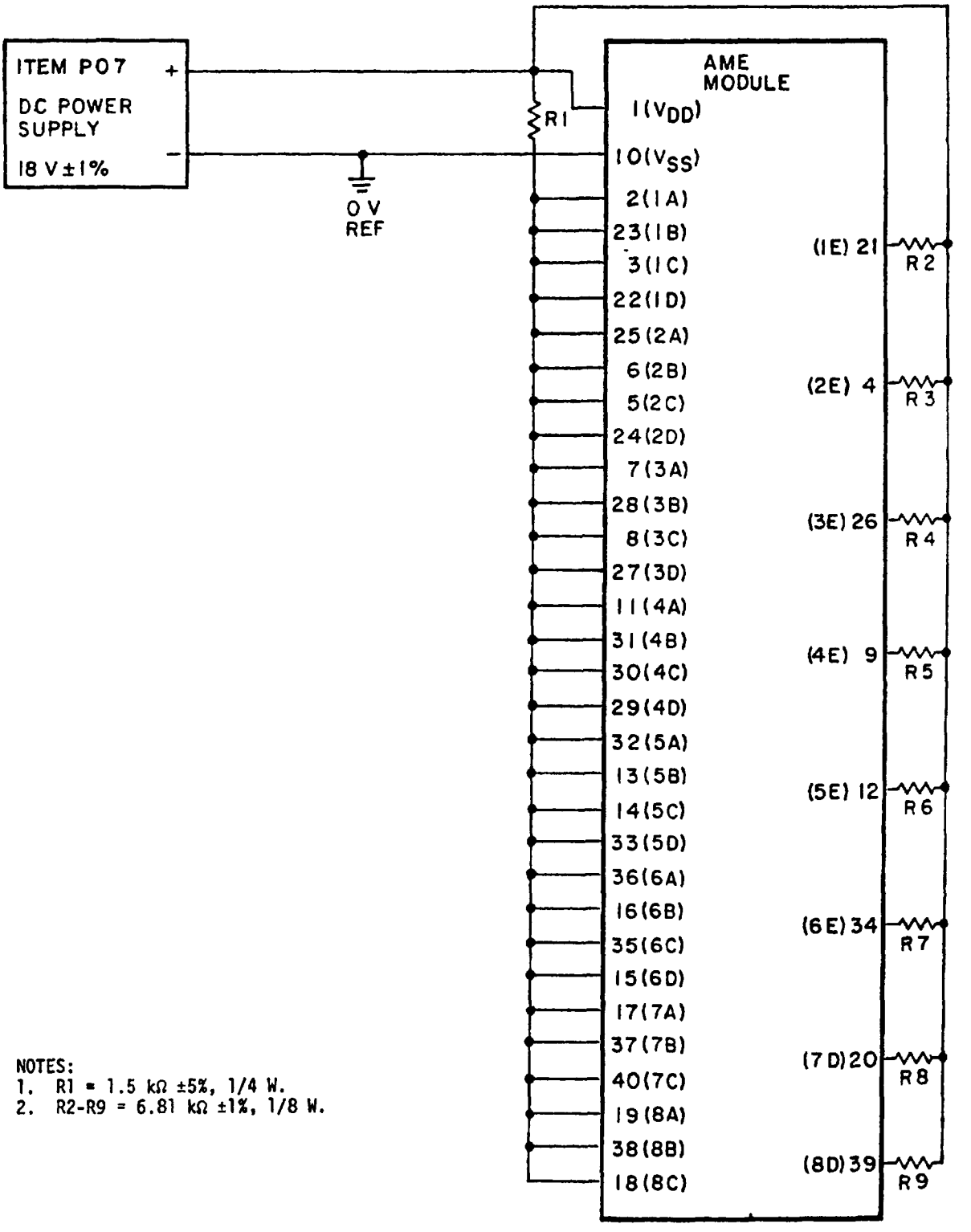
Inputs			Output
A	B	C	D
L	L	L	H
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L



$D = \overline{A+B+C}$

1/ H = High level.
2/ L = Low level.

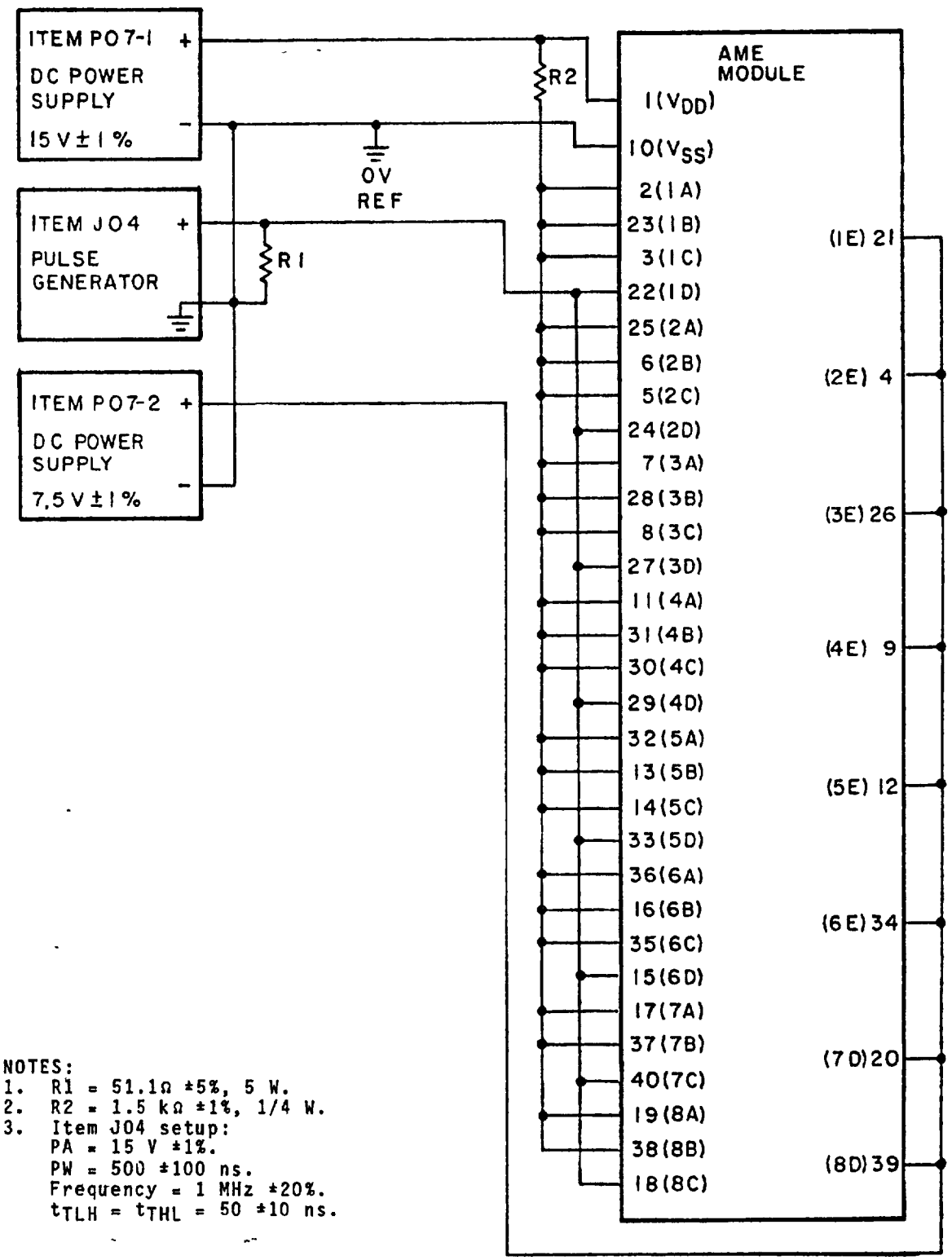
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NOTES:
1. R1 = 1.5 kΩ ±5%, 1/4 W.
2. R2-R9 = 6.81 kΩ ±1%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 51.1Ω ±5%, 5 W.
2. R2 = 1.5 kΩ ±1%, 1/4 W.
3. Item J04 setup:
   PA = 15 V ±1%.
   PW = 500 ±100 ns.
   Frequency = 1 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.

FIGURE 3. Component temperature test circuit.

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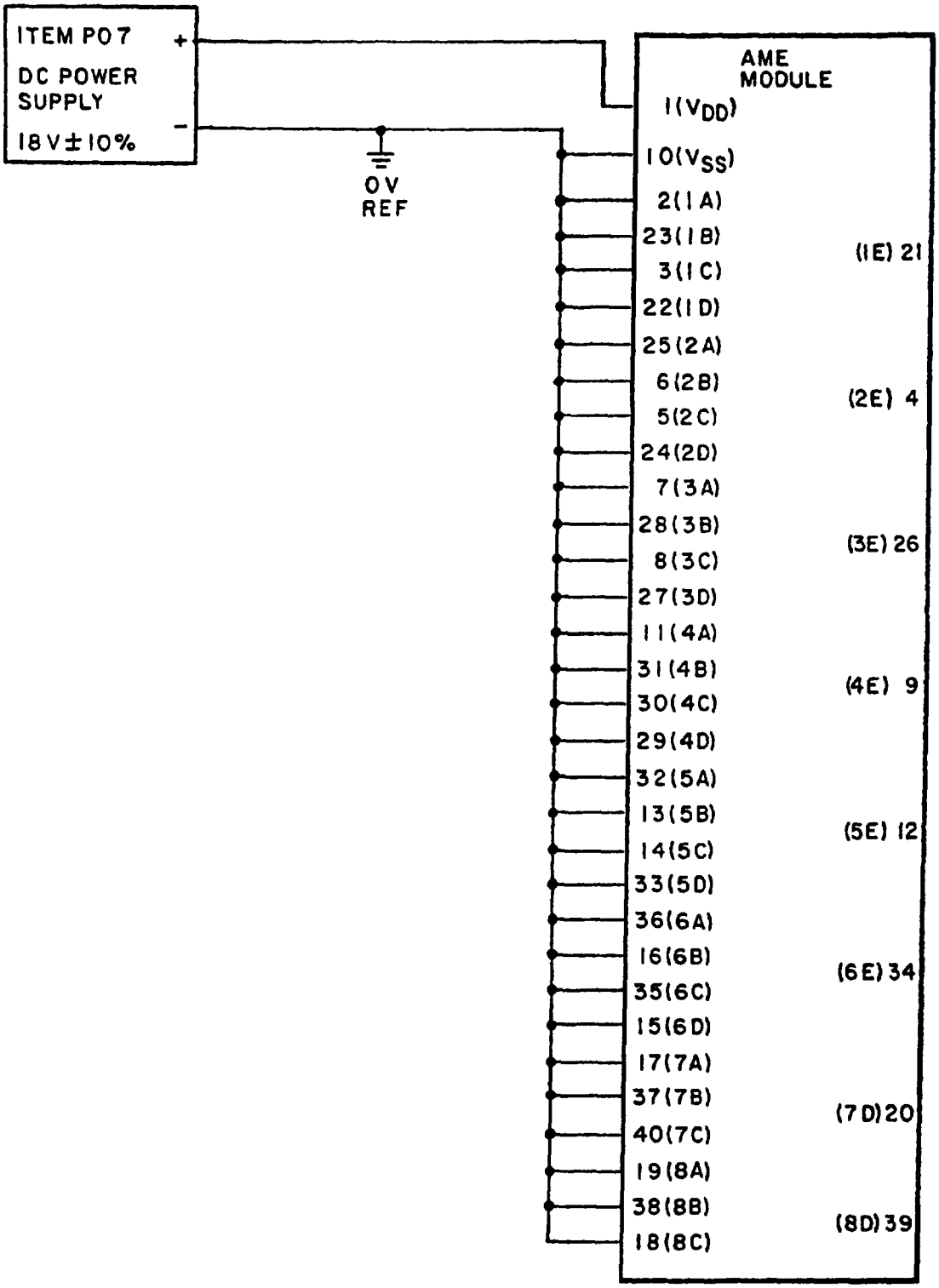


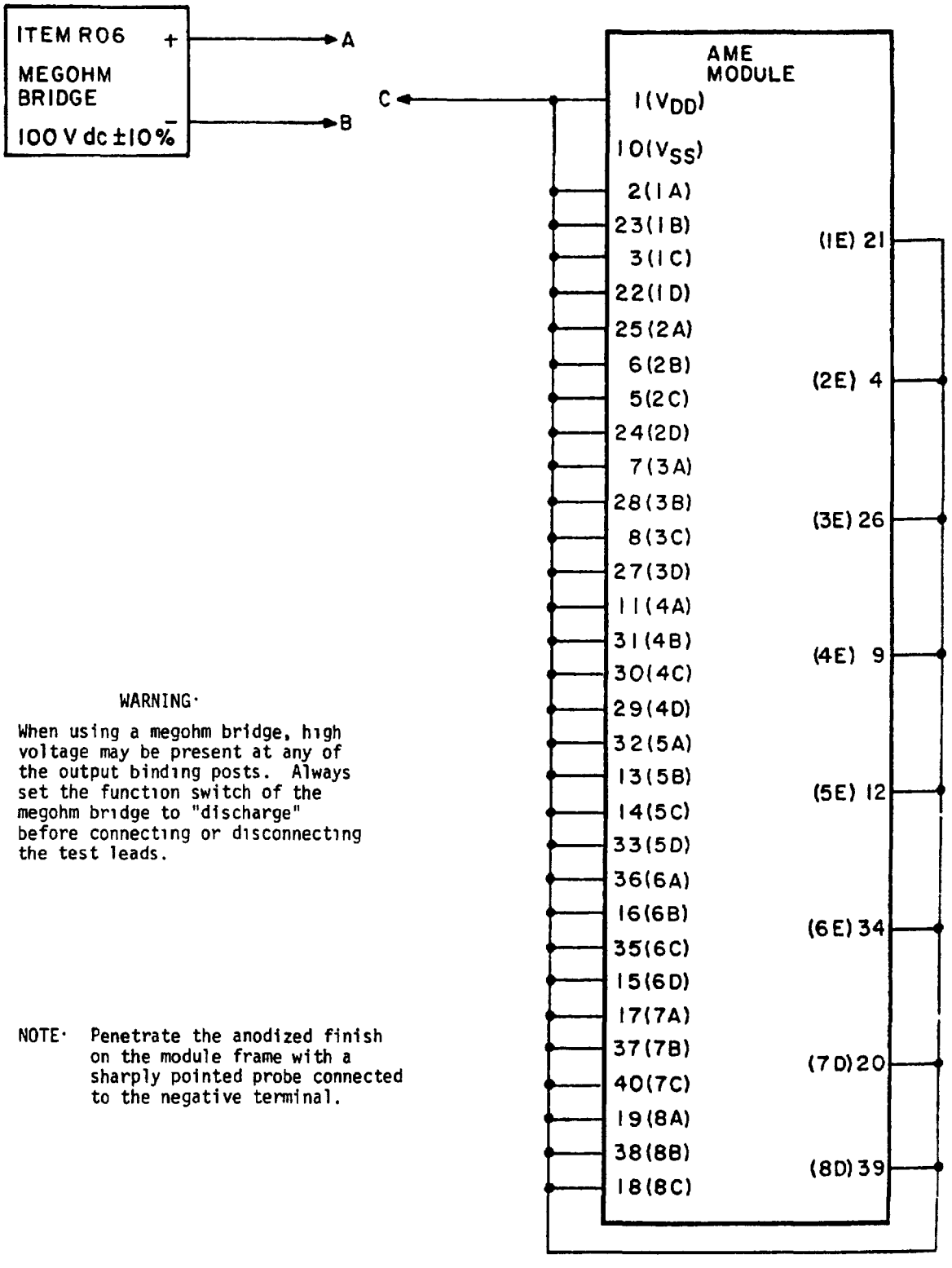
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

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WARNING·

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE· Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Input clamp diode voltage (negative) test.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	1A	Input clamp diode voltage, negative	2	Open	-0.4	-6.0	V	6
B2	1B	"	23	"	"	"	"	"
B3	1C	"	3	"	"	"	"	"
B4	1D	"	22	"	"	"	"	"
B5	2A	"	25	"	"	"	"	"
B6	2B	"	6	"	"	"	"	"
B7	2C	"	5	"	"	"	"	"
B8	2D	"	24	"	"	"	"	"
B9	3A	"	7	"	"	"	"	"
B10	3B	"	28	"	"	"	"	"
B11	3C	"	8	"	"	"	"	"
B12	3D	"	27	"	"	"	"	"
B13	4A	"	11	"	"	"	"	"
B14	4B	"	31	"	"	"	"	"
B15	4C	"	30	"	"	"	"	"
B16	4D	"	29	"	"	"	"	"
B17	5A	"	32	"	"	"	"	"

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TABLE IV. Input clamp diode voltage (negative) test - Continued.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B18	5B	Input clamp diode voltage, negative	13	Open	-0.4	-6.0	V	6
B19	5C	"	14	"	"	"	"	"
B20	5D	"	33	"	"	"	"	"
B21	6A	"	36	"	"	"	"	"
B22	6B	"	16	"	"	"	"	"
B23	6C	"	35	"	"	"	"	"
B24	6D	"	15	"	"	"	"	"
B25	7A	"	17	"	"	"	"	"
B26	7B	"	37	"	"	"	"	"
B27	7C	"	40	"	"	"	"	"
B28	8A	"	19	"	"	"	"	"
B29	8B	"	38	"	"	"	"	"
B30	8C	"	18	"	"	"	"	"

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TABLE V. Input clamp diode voltage (positive) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input clamp diode voltage, positive	Open	2	0.4	1.5	V	6
C2	1B "	"	23	"	"	"	"
C3	1C "	"	3	"	"	"	"
C4	1D "	"	22	"	"	"	"
C5	2A "	"	25	"	"	"	"
C6	2B "	"	6	"	"	"	"
C7	2C "	"	5	"	"	"	"
C8	2D "	"	24	"	"	"	"
C9	3A "	"	7	"	"	"	"
C10	3B "	"	28	"	"	"	"
C11	3C "	"	8	"	"	"	"
C12	3D "	"	27	"	"	"	"
C13	4A "	"	11	"	"	"	"
C14	4B "	"	31	"	"	"	"
C15	4C "	"	30	"	"	"	"
C16	4D "	"	29	"	"	"	"
C17	5A "	"	32	"	"	"	"

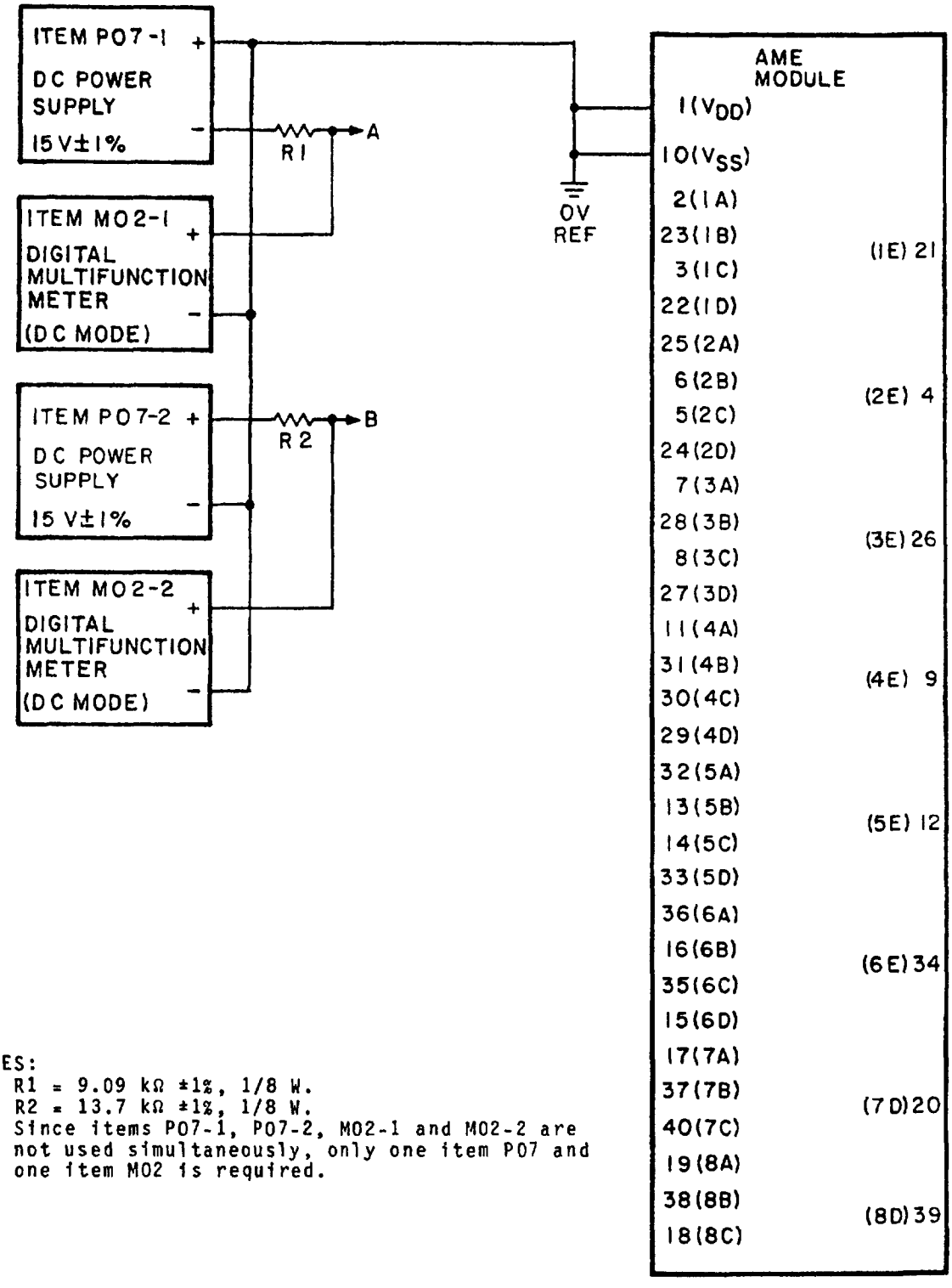
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TABLE V. Input clamp diode voltage (positive) test - Continued.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C18	5B	Input clamp diode voltage, positive	Open	13	0.4	1.5	V	6
C19	5C	"	"	14	"	"	"	"
C20	5D	"	"	33	"	"	"	"
C21	6A	"	"	36	"	"	"	"
C22	6B	"	"	16	"	"	"	"
C23	6C	"	"	35	"	"	"	"
C24	6D	"	"	15	"	"	"	"
C25	7A	"	"	17	"	"	"	"
C26	7B	"	"	37	"	"	"	"
C27	7C	"	"	40	"	"	"	"
C28	8A	"	"	19	"	"	"	"
C29	8B	"	"	38	"	"	"	"
C30	8C	"	"	18	"	"	"	"

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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items PO7-1, PO7-2, MO2-1 and MO2-2 are not used simultaneously, only one item PO7 and one item MO2 is required.

* FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

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TABLE VI. Functional test.

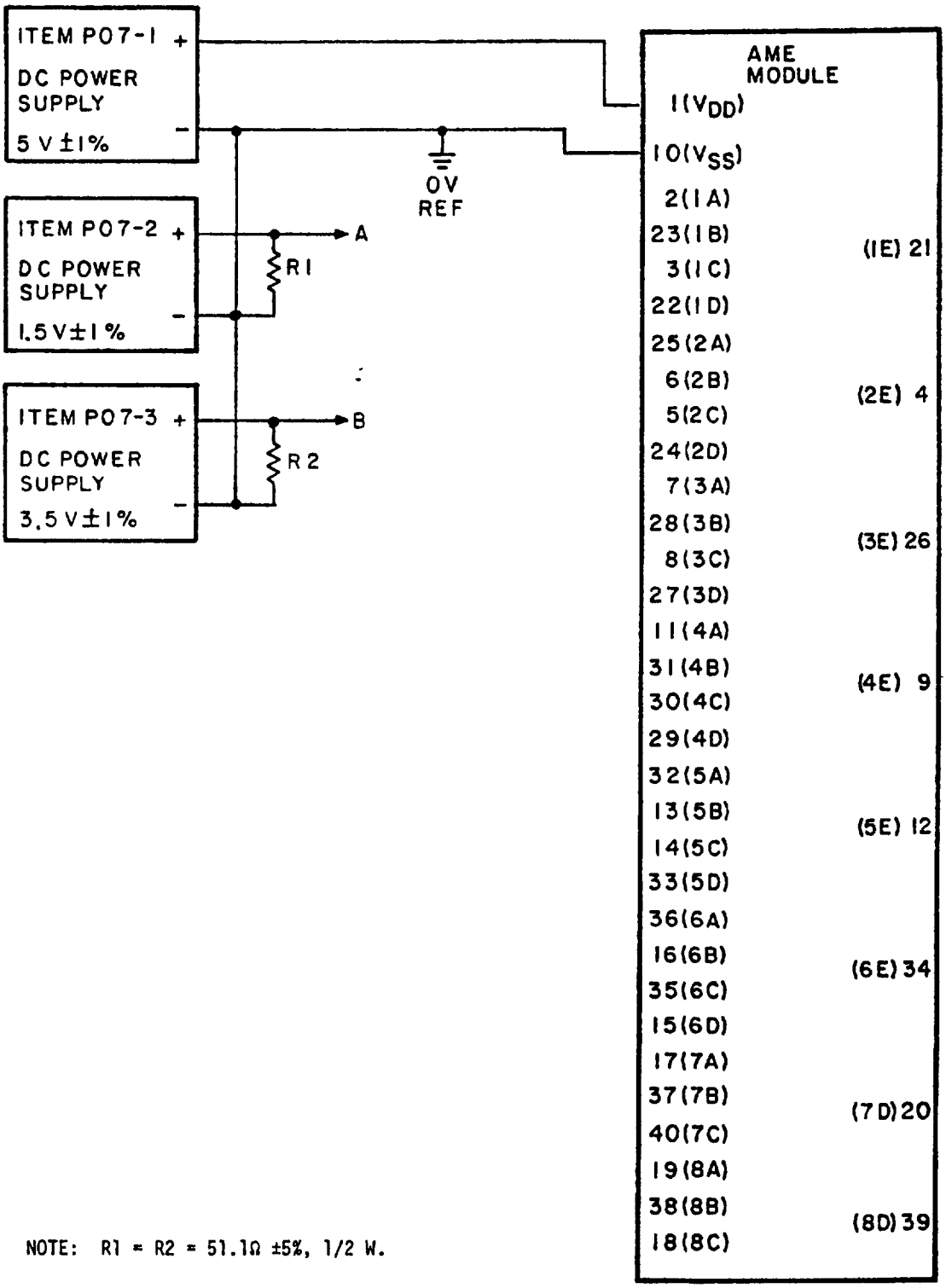
Test condition	Parameter	Connect		Outputs								Test figure
		A to pin	B to pin	Pins 21	4	26	9	12	34	20	39	
D1	Functional	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	Open	H	H	H	H	H	H	H	H	7
D2	"	23,3,22,6,5,24,28,8,27,31,30,29,13,14,33,16,35,15,37,40,38,18	2,25,7,11,32,36,17,19	L	L	L	L	L	L	L	L	"
D3	"	2,3,22,25,5,24,7,8,27,11,30,29,32,14,33,36,35,15,17,40,19,18	23,6,28,31,13,16,37,38	"	"	"	"	"	"	"	"	"
D4	"	3,22,5,24,8,27,30,29,14,33,35,15,40,18	2,23,25,6,7,28,11,31,32,13,36,16,17,37,19,38	"	"	"	"	"	"	"	"	"
D5	"	2,23,22,25,6,24,7,28,27,11,31,29,32,13,33,36,16,15,17,37,19,38	3,5,8,30,14,35,40,18	"	"	"	"	"	"	"	"	"
D6	"	23,22,6,24,28,27,31,29,13,33,16,37,38,35	2,3,25,5,7,8,11,30,32,14,36,15,17,40,19,18	"	"	"	"	"	"	"	"	"
D7	"	2,22,25,24,7,27,11,29,32,33,36,15,17,19	23,3,6,5,28,8,31,30,13,14,16,35,37,40,38,18	"	"	"	"	"	"	"	"	"
D8	"	22,24,27,29,33,15	2,23,3,25,6,5,7,28,8,11,31,30,32,13,14,36,16,35,17,37,40,19,38,18	"	"	"	"	"	"	"	"	"
D9	"	2,23,3,25,6,5,7,28,8,11,31,30,32,13,14,36,16,35,17,37,40,19,38,18	22,24,27,29,33,15	"	"	"	"	"	"	"	"	"
D10	"	23,3,6,5,28,8,31,30,13,14,16,35,17,37,40,19,38,18	2,22,25,24,7,27,11,29,32,33,36,15	"	"	"	"	"	"	"	"	"
D11	"	2,3,25,5,7,8,11,30,32,14,36,35,17,37,40,19,38,18	23,22,6,24,28,27,31,29,13,33,16,15	"	"	"	"	"	"	"	"	"

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TABLE VI. Functional test - Continued.

Test condition	Parameter	Connect		Outputs								Test figure
		A to pin	B to pin	Pins 21	4	26	9	12	34	20	39	
D12	Functional	3,5,8,30,14,35,17,37,40,19,38,18	2,23,22,25,6,24,7,28,27,11,31,29,32,13,33,36,16,15	L	L	L	L	L	L	L	L	7
D13	"	2,23,25,6,7,28,11,31,32,13,36,16,17,37,40,19,38,18	3,22,5,24,8,27,30,29,14,33,35,15	"	"	"	"	"	"	"	"	"
D14	"	23,6,28,31,13,16,17,37,40,19,38,18	2,3,22,25,5,24,7,8,27,11,30,29,32,14,33,36,35,15	"	"	"	"	"	"	"	"	"
D15	"	2,25,7,11,32,36,17,37,40,19,38,18	23,3,22,6,5,24,28,8,27,31,30,29,13,14,33,16,35,15	"	"	"	"	"	"	"	"	"
D16	"	Open	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15	"	"	"	"	"	"	"	"	"

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NOTE: R1 = R2 = 51.1Ω ±5%, 1/2 W.

FIGURE 7. Functional test circuit for table VI.

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*TABLE VII. Output voltage (low-level, I_{OL} = 0) test.

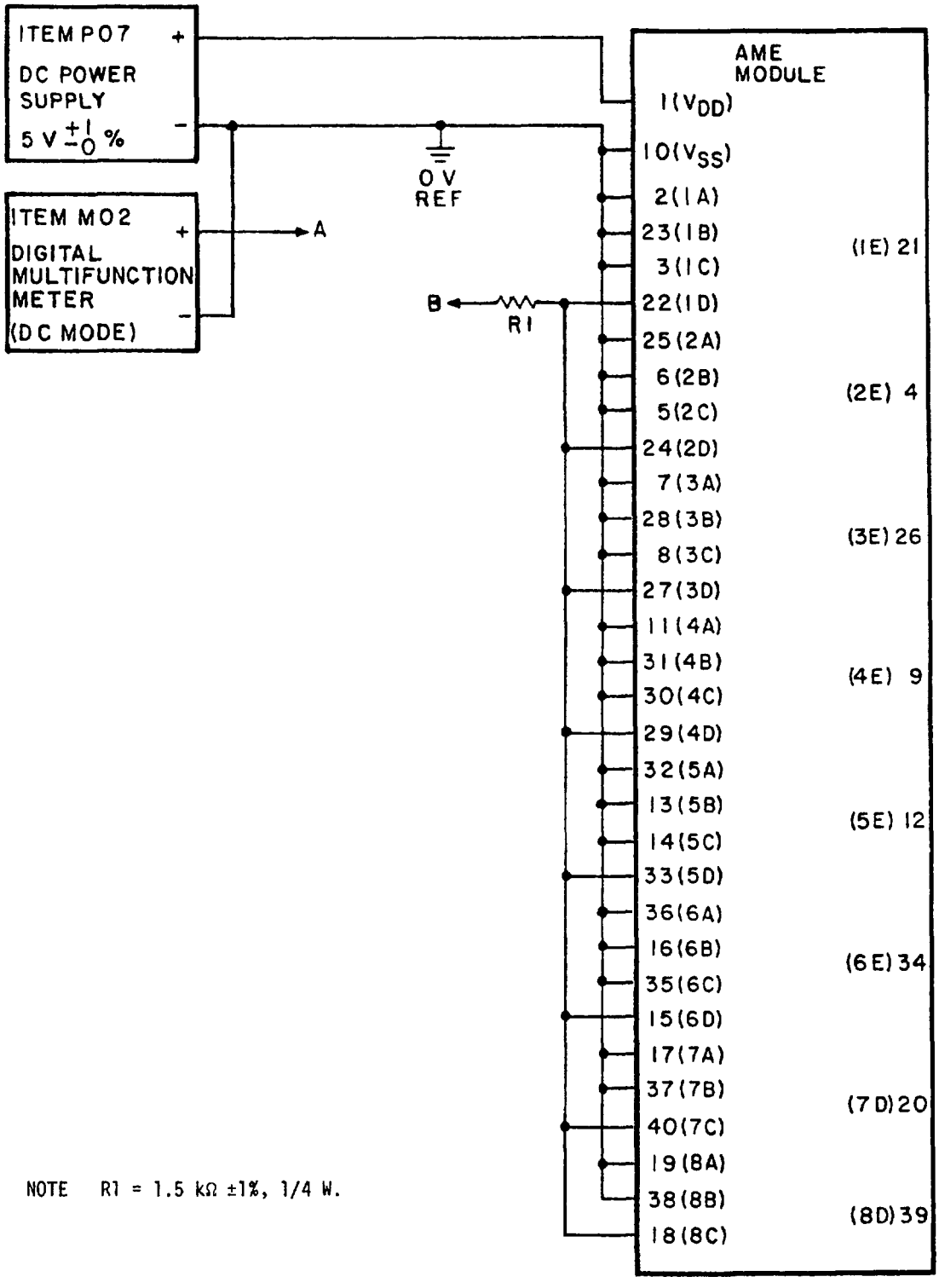
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	1E Output voltage (low level), I_{OL} = 0	21	1	0	50	mV	8
E2	2E "	4	"	"	"	"	"
E3	3E "	26	"	"	"	"	"
E4	4E "	9	"	"	"	"	"
E5	5E "	12	"	"	"	"	"
E6	6E "	34	"	"	"	"	"
E7	7D "	20	"	"	"	"	"
E8	8D "	39	"	"	"	"	"

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*TABLE VIII. Output voltage (high-level, I_{OH} = 0) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	1E Output voltage (high level), I_{OH} = 0	21	10	4.95	5.05	V	8
F2	2E "	4	"	"	"	"	"
F3	3E "	26	"	"	"	"	"
F4	4E "	9	"	"	"	"	"
F5	5E "	12	"	"	"	"	"
F6	6E "	34	"	"	"	"	"
F7	7D "	20	"	"	"	"	"
F8	8D "	39	"	"	"	"	"

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NOTE R1 = 1.5 kΩ ±1%, 1/4 W.

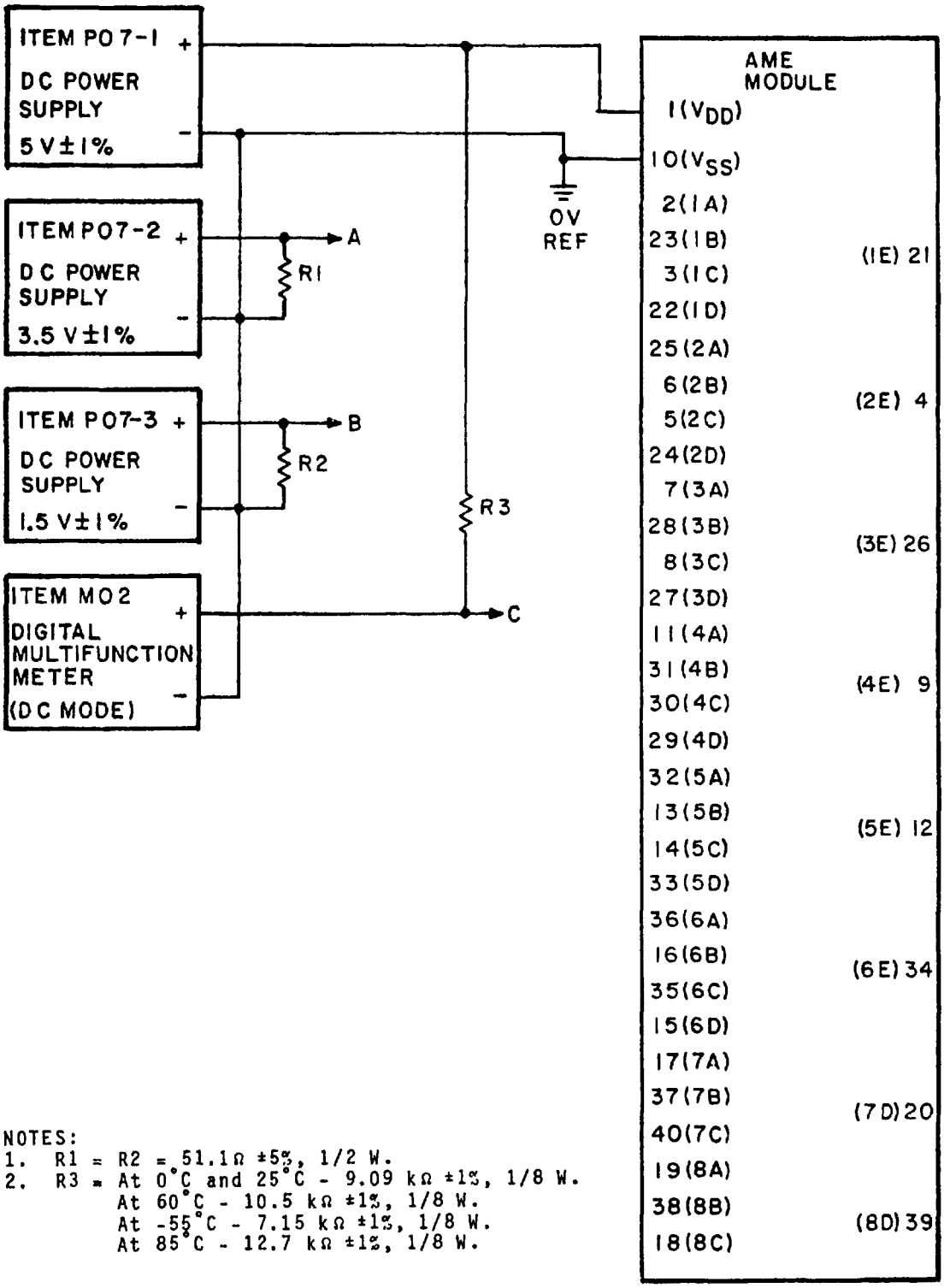
* FIGURE 8. Output voltage (low-level, I_{OL} = 0 and high-level, I_{OH} = 0) test circuit for tables VII and VIII.

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*TABLE IX. Output voltage (low-level, V_{DD} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1E Output voltage (low level), V_{DD} = 5 V	2,23,3,22	25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	21	0	400	mV	9
G2	2E "	25,6,5,24	2,23,3,22,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	4	"	"	"	"
G3	3E "	7,28,8,27	2,23,3,22,25,6,5,24,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	26	"	"	"	"
G4	4E "	11,31,30,29	2,23,3,22,25,6,5,24,7,28,8,27,32,13,14,33,36,16,35,15,17,37,40,19,38,18	9	"	"	"	"
G5	5E "	32,13,14,33	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,36,16,35,15,17,37,40,19,38,18	12	"	"	"	"
G6	6E "	36,16,35,15	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,17,37,40,19,38,18	34	"	"	"	"
G7	7D "	17,37,40	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,19,38,18	20	"	"	"	"
G8	8D "	19,38,18	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40	39	"	"	"	"

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = At 0°C and 25°C - 9.09 kΩ ±1%, 1/8 W.
At 60°C - 10.5 kΩ ±1%, 1/8 W.
At -55°C - 7.15 kΩ ±1%, 1/8 W.
At 85°C - 12.7 kΩ ±1%, 1/8 W.

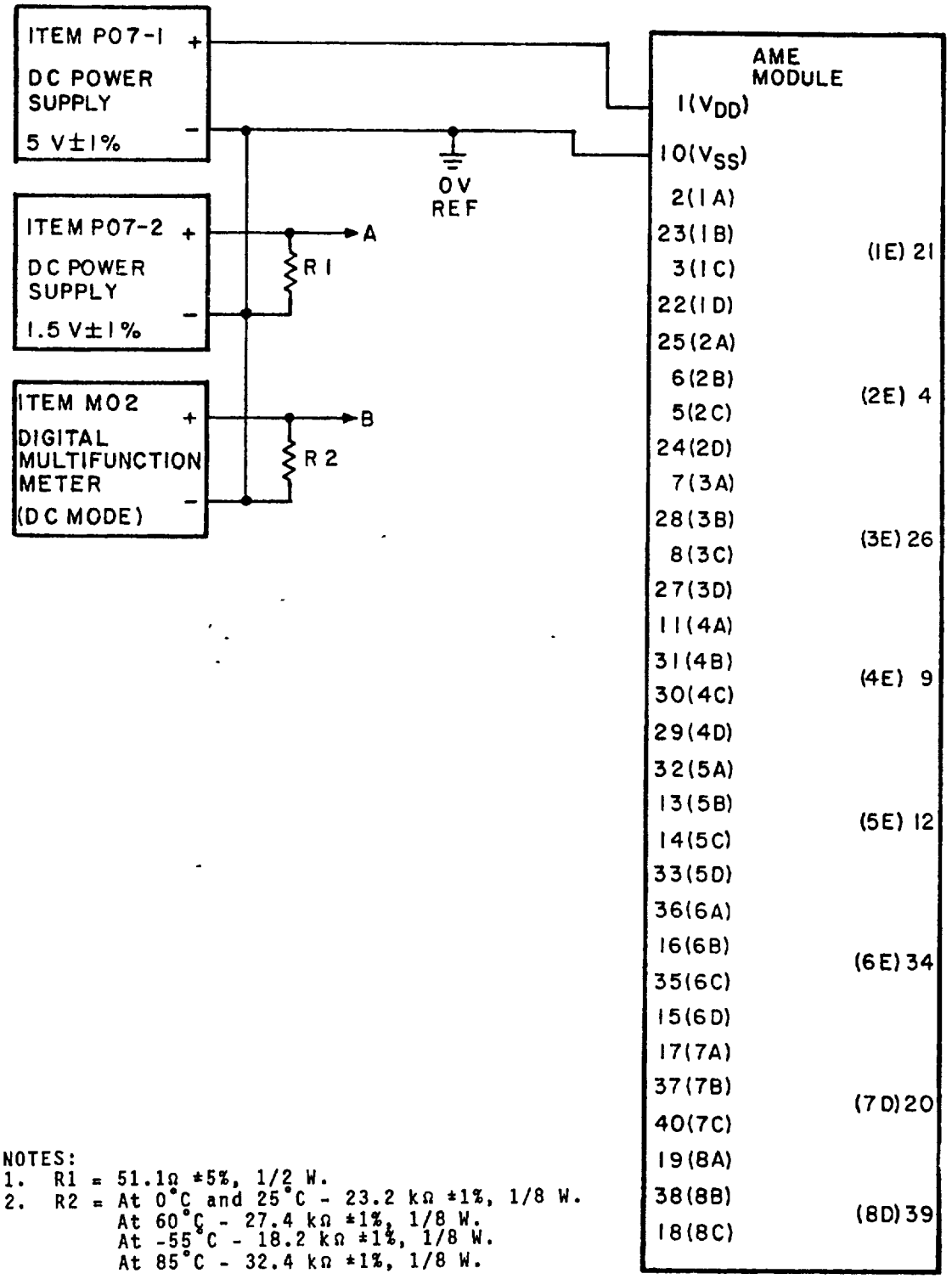
* FIGURE 9. Output voltage (low-level, V_{DD} = 5 V) test circuit for table IX.

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*TABLE X. Output voltage (high-level, V_{DD} = 5 V) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
H1	1E Output voltage (high level), V_{DD} = 5 V	23,3,22,25,6, 5,24,7,28,8,27, 11,31,30,29,32,13,14,33,36, 16,35,15,17,37,40,19,38,18,2	21	4.6	5.0	V	10
H2	2E "	2,23,3,22,6,5,24,7,28,8,27, 11,31,30,29,32,13,14,33,36, 16,35,15,17,37,40,19,38,18,25	4	"	"	"	"
H3	3E "	2,23,3,22,25,6,5,24,28,8,27, 11,31,30,29,32,13,14,33,36, 16,35,15,17,37,40,19,38,18,7	26	"	"	"	"
H4	4E "	2,23,3,22,25,6,5,24,7,28,8, 27,31,30,29,32,13,14,33,16, 35,15,17,37,40,19,18,38,11	9	"	"	"	"
H5	5E "	2,23,3,22,25,6,5,24,7,28,8, 27,11,31,30,29,13,14,33,36, 16,35,15,17,37,40,19,38,18, 32	12	"	"	"	"
H6	6E "	2,23,3,22,25,6,5,24,7,28,8, 27,11,31,30,29,32,13,14,33, 16,35,15,17,37,40,19,38,18,36	34	"	"	"	"
H7	7D "	2,23,3,22,25,6,5,24,7,28,8, 27,11,31,30,29,32,13,14,33, 36,16,35,15,37,40,19,38,18,17	20	"	"	"	"
H8	8D "	2,23,3,22,25,6,5,24,7,28,8, 27,11,31,30,29,32,13,14,33, 36,16,35,15,17,37,40,38,18,19	39	"	"	"	"

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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = At 0°C and 25°C - 23.2 kΩ ±1%, 1/8 W.
   At 60°C - 27.4 kΩ ±1%, 1/8 W.
   At -55°C - 18.2 kΩ ±1%, 1/8 W.
   At 85°C - 32.4 kΩ ±1%, 1/8 W.

* FIGURE 10. Output voltage (high-level, V_{DD} = 5 V) test circuit for table X.

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*TABLE XI. Output voltage (low-level, V_{DD} = 15 V) test.

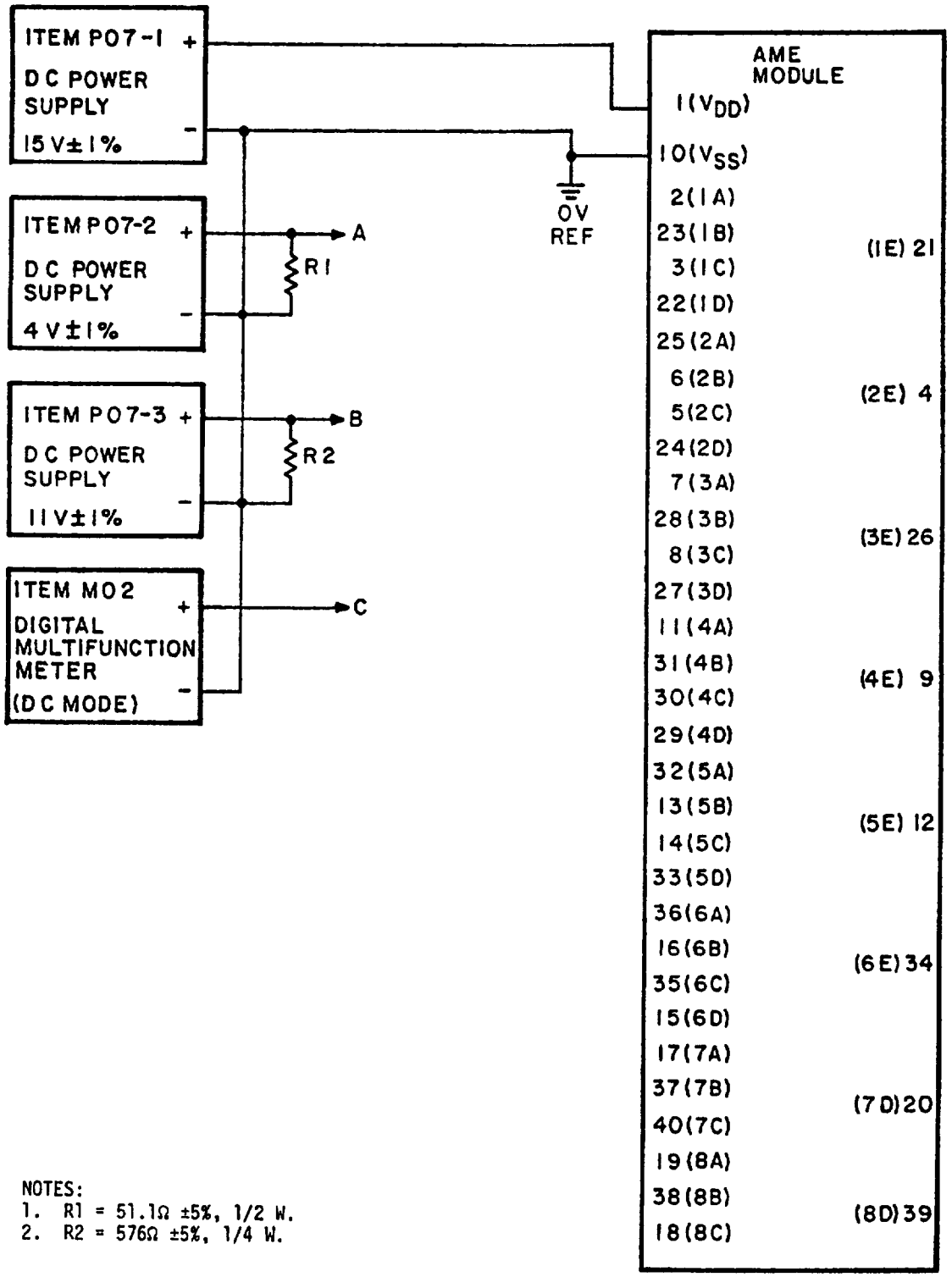
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1E Output voltage (low level), V_{DD} = 15 V	25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2,23,3,22	21	0.0	1.5	V	11
J2	2E "	2,23,3,22,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25,6,5,24	4	"	"	"	"
J3	3E "	2,23,3,22,25,6,5,24,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7,28,8,27	26	"	"	"	"
J4	4E "	2,23,3,22,25,6,5,24,7,28,8,27,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11,31,30,29	9	"	"	"	"
J5	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,36,16,35,15,17,37,40,19,38,18	32,13,14,33	12	"	"	"	"
J6	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,17,37,40,19,38,18	36,16,35,15	34	"	"	"	"
J7	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,19,38,18	17,37,40	20	"	"	"	"
J8	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40	19,38,18	39	"	"	"	"

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*TABLE XII. Output voltage (high-level, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1E Output voltage (high level), V_{DD} = 15 V	2,23,3,22	25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	21	13.5	15.0	V	11
K2	2E "	25,6,5,24	2,23,3,22,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	4	"	"	"	"
K3	3E "	7,28,8,27	2,23,3,22,25,6,5,24,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	26	"	"	"	"
K4	4E "	11,31,30,29	2,23,3,22,25,6,5,24,7,28,8,27,32,13,14,33,36,16,35,15,17,37,40,19,38,18	9	"	"	"	"
K5	5E "	32,13,14,33	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,36,16,35,15,17,37,40,19,38,18	12	"	"	"	"
K6	6E "	36,16,35,15	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,17,37,40,19,38,18	34	"	"	"	"
K7	7D "	17,37,40	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,19,38,18	20	"	"	"	"
K8	8D "	19,38,18	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40	39	"	"	"	"

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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = 576Ω ±5%, 1/4 W.

* FIGURE 11. Output voltage (low-level and high-level, V_{DD} = 15 V) test circuit for tables XI and XII.

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TABLE XIII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	1A Input current, high level	2	23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	0	200	nA	12
L2	1B "	23	2,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L3	1C "	3	2,23,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	"	"	"	"
L4	1D "	22	2,23,3,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L5	2A "	25	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L6	2B "	6	2,23,3,22,25,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	"	"	"	"
L7	2C "	5	2,23,3,22,25,6,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	"	"	"	"
L8	2D "	24	2,23,3,22,25,5,6,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	"	"	"	"
L9	3A "	7	2,23,3,22,25,6,5,24,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	"	"	"	"
L10	3B "	28	2,23,3,22,25,6,5,24,7,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	"	"	"	"

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L11	3C Input current, high level	8	2,23,3,22,25,6,5,24,7,28,27,11, 31,30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	0	200	nA	12
L12	3D "	27	2,23,3,22,25,6,5,24,7,28,8,11,31, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L13	4A "	11	2,23,3,22,25,6,5,24,7,28,8,27,31, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L14	4B "	31	2,23,3,22,25,6,5,24,7,28,8,27,11, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L15	4C "	30	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L16	4D "	29	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L17	5A "	32	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L18	5B "	13	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L19	5C "	14	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
L20	5D "	33	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,36,16,35,15,17, 37,40,19,38,18	"	"	"	"

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L21	6A Input current, high level	36	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	0	200	nA	12
L22	6B "	16	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,35,15,17,37,40,19,38,18	"	"	"	"
L23	6C "	35	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,15,17,37,40,19,38,18	"	"	"	"
L24	6D "	15	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,17,37,40,19,38,18	"	"	"	"
L25	7A "	17	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,37,40,19,38,18	"	"	"	"
L26	7B "	37	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,40,19,38,18	"	"	"	"
L27	7C "	40	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,19,38,18	"	"	"	"
L28	8A "	19	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	"	"	"	"
L29	8B "	38	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,18	"	"	"	"
L30	8C "	18	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:02Z
Check the source to verify that this is the current version before use.

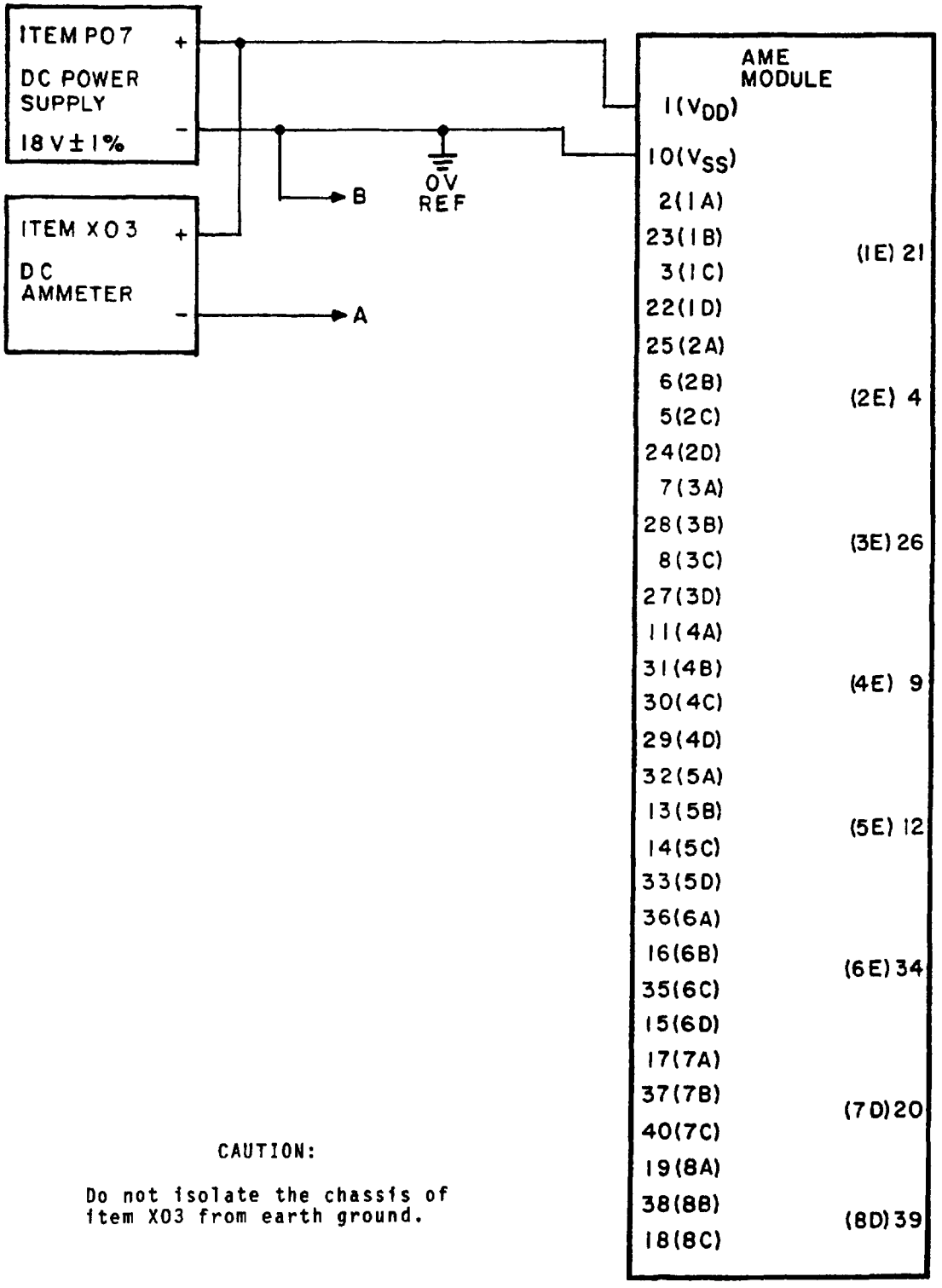


FIGURE 12. Input current (high-level) test circuit for table XIII.

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TABLE XIV. Input current (low-level) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
M1	1A Input current, low level	2	23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	0	-200	nA	13
M2	1B "	23	2,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M3	1C "	3	2,23,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	"	"	"	"
M4	1D "	22	2,23,3,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M5	2A "	25	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M6	2B "	6	2,23,3,22,25,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	"	"	"	"
M7	2C "	5	2,23,3,22,25,6,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	"	"	"	"
M8	2D "	24	2,23,3,22,25,5,6,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	"	"	"	"
M9	3A "	7	2,23,3,22,25,6,5,24,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	"	"	"	"
M10	3B "	28	2,23,3,22,25,6,5,24,7,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	"	"	"	"

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TABLE XIV. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M11	3C Input current, low level	8	2,23,3,22,25,6,5,24,7,28,27,11, 31,30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	0	-200	nA	13
M12	3D "	27	2,23,3,22,25,6,5,24,7,28,8,11,31, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M13	4A "	11	2,23,3,22,25,6,5,24,7,28,8,27,31, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M14	4B "	31	2,23,3,22,25,6,5,24,7,28,8,27,11, 30,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M15	4C "	30	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,29,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M16	4D "	29	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,32,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M17	5A "	32	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,13,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M18	5B "	13	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,14,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M19	5C "	14	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,33,36,16,35,15,17, 37,40,19,38,18	"	"	"	"
M20	5D "	33	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,36,16,35,15,17, 37,40,19,38,18	"	"	"	"

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TABLE XIV. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M21	6A Input current, low level	36	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	0	-200	nA	13
M22	6B "	16	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,35,15,17,37,40,19,38,18	"	"	"	"
M23	6C "	35	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,15,17,37,40,19,38,18	"	"	"	"
M24	6D "	15	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,17,37,40,19,38,18	"	"	"	"
M25	7A "	17	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,37,40,19,38,18	"	"	"	"
M26	7B "	37	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,40,19,38,18	"	"	"	"
M27	7C "	40	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,19,38,18	"	"	"	"
M28	8A "	19	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	"	"	"	"
M29	8B "	38	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,18	"	"	"	"
M30	8C "	18	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38	"	"	"	"

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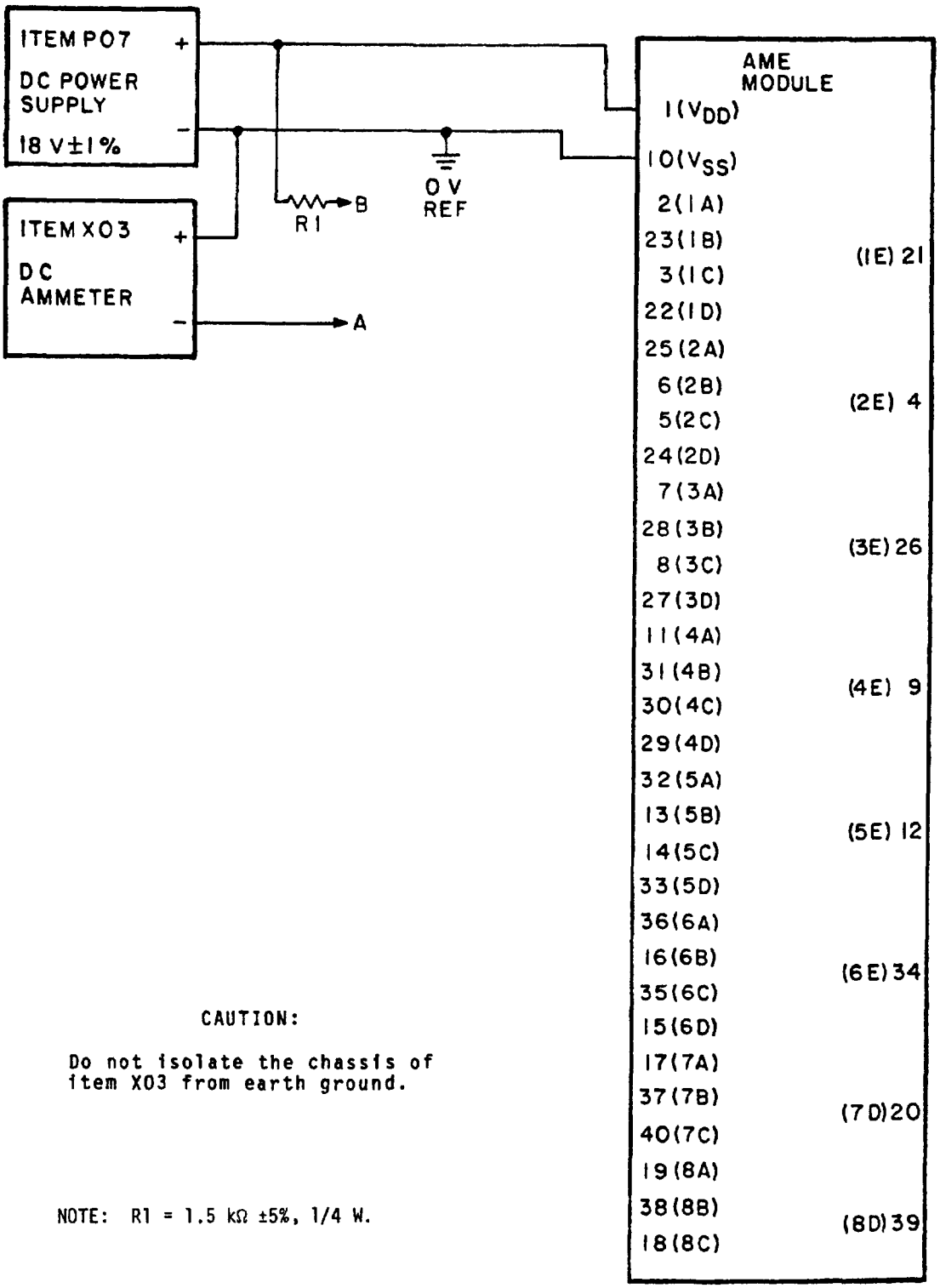


FIGURE 13. Input current (low-level) test circuit for table XIV.

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TABLE XV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Power supply current, low level	Open	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	0	8	µA	14
N2	Power supply current, high level	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	Open	0	8	µA	14

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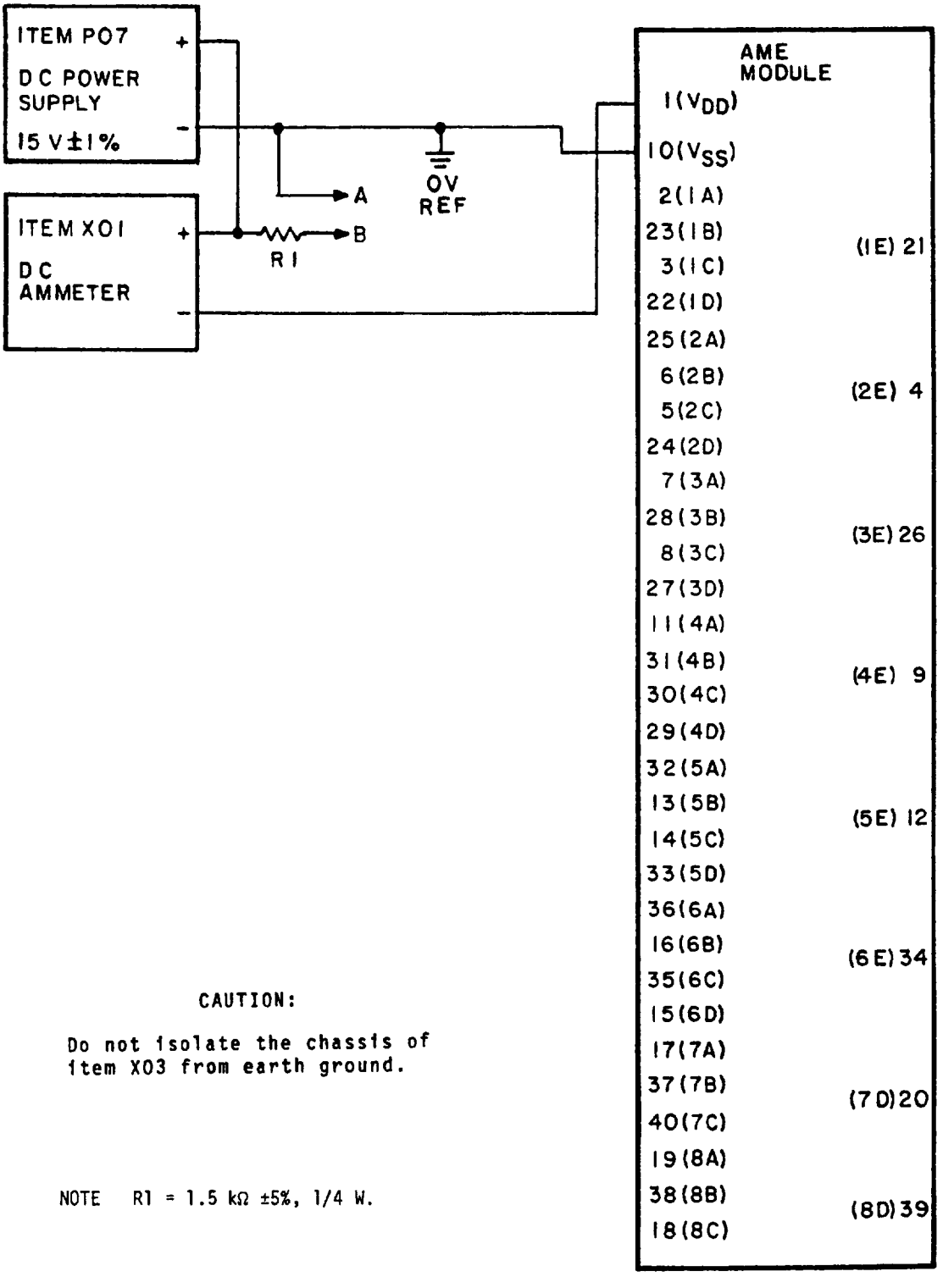


FIGURE 14. Power supply current (low-level and high-level) test circuit for table XV.

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TABLE XVI. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
P1	1E Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	2	21	20	200	ns	15,16
P2	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,19,38,18	25	4	"	"	"	"
P3	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,19,38,18	7	26	"	"	"	"
P4	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30, 29,32,13,14,33,36,16,35,15,17,37,40, 19,38,18	11	9	"	"	"	"
P5	5E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30, 29,13,14,33,36,16,35,15,17,37,40,19, 38,18,11	32	12	"	"	"	"
P6	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,16,35,15,17,37,40,19, 38,18	36	34	"	"	"	"
P7	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15 37,40, 19,38,18	17	20	"	"	"	"
P8	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,38,18	19	39	"	"	"	"
P9	1E Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	2	21	68	680	"	"
P10	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31,30, 29,32,13,14,33,36,16,35,15,17,37,40, 19,38,18	25	4	"	"	"	"
P11	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,19,38,18	7	26	"	"	"	"

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TABLE XVI. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to	Min	Max		
P12	4E Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9	68	680	ns	15,16
P13	5E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18,11	32	12	"	"	"	"
P14	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	36	34	"	"	"	"
P15	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15 37,40,19,38,18	17	20	"	"	"	"
P16	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	19	39	"	"	"	"

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TABLE XVII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
Q1	1E Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	2	21	20	200	ns	15,16
Q2	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,19,38,18	25	4	"	"	"	"
Q3	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,19,38,18	7	26	"	"	"	"
Q4	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30, 29,32,13,14,33,36,16,35,15,17,37,40, 19,38,18	11	9	"	"	"	"
Q5	5E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30, 29,13,14,33,36,16,35,15,17,37,40,19, 38,18,11	32	12	"	"	"	"
Q6	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,16,35,15,17,37,40, 19,38,18	36	34	"	"	"	"
Q7	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15 37,40, 19,38,18	17	20	"	"	"	"
Q8	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,38,18	19	39	"	"	"	"
Q9	1E Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	2	21	68	680	"	"
Q10	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31,30, 29,32,13,14,33,36,16,35,15,17,37,40, 19,38,18	25	4	"	"	"	"
Q11	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,19,38,18	7	26	"	"	"	"

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TABLE XVII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q12	4E Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9	68	680	ns	15,16
Q13	5E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18,11	32	12	"	"	"	"
Q14	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,16,35,15,17,37,40,19,38,18	36	34	"	"	"	"
Q15	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15 37,40,19,38,18	17	20	"	"	"	"
Q16	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	19	39	"	"	"	"

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TABLE XVIII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	1E Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	10	100	ns	15,16
R2	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4	"	"	"	"
R3	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26	"	"	"	"
R4	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9	"	"	"	"
R5	5E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18,11	32	12	"	"	"	"
R6	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,16,35,15,17,37,40,19,38,18	36	34	"	"	"	"
R7	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15 37,40,19,38,18	17	20	"	"	"	"
R8	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	19	39	"	"	"	"

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TABLE XIX. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	1E Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	2	21	10	100	ns	15,16
S2	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,19,38,18	25	4	"	"	"	"
S3	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,19,38,18	7	26	"	"	"	"
S4	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30, 29,32,13,14,33,36,16,35,15,17,37,40, 19,38,18	11	9	"	"	"	"
S5	5E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30, 29,13,14,33,36,16,35,15,17,37,40,19, 38,18,11	32	12	"	"	"	"
S6	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,16,35,15,17,37,40, 19,38,18	36	34	"	"	"	"
S7	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15 37,40, 19,38,18	17	20	"	"	"	"
S8	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,38,18	19	39	"	"	"	"

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TABLE XX. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	1E Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38,18	2	21	8	80	ns	15,16
T2	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,19,38,18	25	4	"	"	"	"
T3	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,19,38,18	7	26	"	"	"	"
T4	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30, 29,32,13,14,33,36,16,35,15,17,37,40, 19,38,18	11	9	"	"	"	"
T5	5E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30, 29,13,14,33,36,16,35,15,17,37,40,19, 38,18,11	32	12	"	"	"	"
T6	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,16,35,15,17,37,40,19, 38,18	36	34	"	"	"	"
T7	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15 37,40, 19,38,18	17	20	"	"	"	"
T8	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15,17,37, 40,38,18	19	39	"	"	"	"

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TABLE XXI. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U1	1E Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	8	80	ns	15,16
U2	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4	"	"	"	"
U3	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26	"	"	"	"
U4	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9	"	"	"	"
U5	5E "	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18,11	32	12	"	"	"	"
U6	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	36	34	"	"	"	"
U7	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15 37,40,19,38,18	17	20	"	"	"	"
U8	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	19	39	"	"	"	"

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TABLE XXII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V1	1E Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	2	21	32	320	ns	15,16
V2	1E "	2,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	23	21	"	"	"	"
V3	1E "	2,23,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	3	21	"	"	"	"
V4	1E "	2,23,3,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	22	21	"	"	"	"
V5	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	25	4	"	"	"	"
V6	2E "	2,23,3,22,25,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	6	4	"	"	"	"
V7	2E "	2,23,3,22,25,6,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	5	4	"	"	"	"
V8	2E "	2,23,3,22,25,6,5,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	24	4	"	"	"	"
V9	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	7	26	"	"	"	"
V10	3E "	2,23,3,22,25,6,5,24,7,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	28	26	"	"	"	"

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TABLE XXII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V11	3E Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	2,23,3,22,25,6,5,24,7,28,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	8	26	32	320	ns	15,16
V12	3E "	2,23,3,22,25,6,5,24,7,28,8,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	27	26	"	"	"	"
V13	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	11	9	"	"	"	"
V14	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	31	9	"	"	"	"
V15	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	30	9	"	"	"	"
V16	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	29	9	"	"	"	"
V17	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,13,14,33,36,16,35,15, 17,37,40,19,38,18	32	12	"	"	"	"
V18	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,14,33,36,16,35,15, 17,37,40,19,38,18	32	12	"	"	"	"
V19	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,33,36,16,35,15, 17,37,40,19,38,18	14	12	"	"	"	"
V20	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,36,16,35,15, 17,37,40,19,38,18	33	12	"	"	"	"

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TABLE XXII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V21	6E Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,16,35,15, 17,37,40,19,38,18	36	34	32	320	ns	15,16
V22	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,35,15, 17,37,40,19,38,18	16	34	"	"	"	"
V23	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,15, 17,37,40,19,38,18	35	34	"	"	"	"
V24	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 17,37,40,19,38,18	15	34	"	"	"	"
V25	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,37,40,19,38,18	17	20	"	"	"	"
V26	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,40,19,38,18	37	20	"	"	"	"
V27	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,19,38,18	40	20	"	"	"	"
V28	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,38,18	19	39	"	"	"	"
V29	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,18	38	39	"	"	"	"
V30	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38	18	39	"	"	"	"

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TABLE XXII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
V31	1E Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	2	21	52	520	ns	15,16
V32	1E "	2,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	23	21	"	"	"	"
V33	1E "	2,23,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	3	21	"	"	"	"
V34	1E "	2,23,3,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	22	21	"	"	"	"
V35	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	25	4	"	"	"	"
V36	2E "	2,23,3,22,25,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	6	4	"	"	"	"
V37	2E "	2,23,3,22,25,6,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	5	4	"	"	"	"
V38	2E "	2,23,3,22,25,6,5,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	24	4	"	"	"	"
V39	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	7	26	"	"	"	"
V40	3E "	2,23,3,22,25,6,5,24,7,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	28	26	"	"	"	"

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TABLE XXII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
V41	3E	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	2,23,3,22,25,6,5,24,7,28,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	8	26	52	520	ns	15,16
V42	3E	"	2,23,3,22,25,6,5,24,7,28,8,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	27	26	"	"	"	"
V43	4E	"	2,23,3,22,25,6,5,24,7,28,8,27,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	11	9	"	"	"	"
V44	4E	"	2,23,3,22,25,6,5,24,7,28,8,27,11, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	31	9	"	"	"	"
V45	4E	"	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	30	9	"	"	"	"
V46	4E	"	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	29	9	"	"	"	"
V47	5E	"	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,13,14,33,36,16,35,15, 17,37,40,19,38,18	32	12	"	"	"	"
V48	5E	"	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,14,33,36,16,35,15, 17,37,40,19,38,18	13	12	"	"	"	"
V49	5E	"	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,33,36,16,35,15, 17,37,40,19,38,18	14	12	"	"	"	"
V50	5E	"	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,36,16,35,15, 17,37,40,19,38,18	33	12	"	"	"	"

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TABLE XXII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V51	6E Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,16,35,15, 17,37,40,19,38,18	36	34	52	520	ns	15,16
V52	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,35,15, 17,37,40,19,38,18	16	34	"	"	"	"
V53	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,15, 17,37,40,19,38,18	35	34	"	"	"	"
V54	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 17,37,40,19,38,18	15	34	"	"	"	"
V55	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,37,40,19,38,18	17	20	"	"	"	"
V56	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,37,16,35, 15,17,40,19,38,18	37	20	"	"	"	"
V57	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,37,16,35, 15,17,37,19,38,18	40	20	"	"	"	"
V58	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,37,16,35, 15,17,37,40,38,18	19	39	"	"	"	"
V59	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,37,16,35, 15,17,37,40,19,18	38	39	"	"	"	"
V60	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,37,16,35, 15,17,37,40,19,38	18	39	"	"	"	"

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TABLE XXIII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W1	1E Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	2	21	32	320	ns	15,16
W2	1E "	2,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	23	21	"	"	"	"
W3	1E "	2,23,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	3	21	"	"	"	"
W4	1E "	2,23,3,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	22	21	"	"	"	"
W5	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	25	4	"	"	"	"
W6	2E "	2,23,3,22,25,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	6	4	"	"	"	"
W7	2E "	2,23,3,22,25,6,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	5	4	"	"	"	"
W8	2E "	2,23,3,22,25,6,5,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	24	4	"	"	"	"
W9	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	7	26	"	"	"	"
W10	3E "	2,23,3,22,25,6,5,24,7,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	28	26	"	"	"	"

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TABLE XXIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W11	3E Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	2,23,3,22,25,6,5,24,7,28,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	8	26	32	320	ns	15,16
W12	3E "	2,23,3,22,25,6,5,24,7,28,8,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	27	26	"	"	"	"
W13	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	11	9	"	"	"	"
W14	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	31	9	"	"	"	"
W15	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	30	9	"	"	"	"
W16	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	29	9	"	"	"	"
W17	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,13,14,33,36,16,35,15, 17,37,40,19,38,18	32	12	"	"	"	"
W18	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,14,33,36,16,35,15, 17,37,40,19,38,18	13	12	"	"	"	"
W19	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,33,36,16,35,15, 17,37,40,19,38,18	14	12	"	"	"	"
W20	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,36,16,35,15, 17,37,40,19,38,18	33	12	"	"	"	"

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TABLE XXIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
W21	6E Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,16,35,15, 17,37,40,19,38,18	36	34	32	320	ns	15,16
W22	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,35,15, 17,37,40,19,38,18	16	34	"	"	"	"
W23	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,15, 17,37,40,19,38,18	35	34	"	"	"	"
W24	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 17,37,40,19,38,18	15	34	"	"	"	"
W25	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,37,40,19,38,18	17	20	"	"	"	"
W26	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,37,16,35, 15,17,40,19,38,18	37	20	"	"	"	"
W27	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,37,16,35, 15,17,37,19,38,18	40	20	"	"	"	"
W28	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,37,16,35, 15,17,37,40,38,18	19	39	"	"	"	"
W29	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,37,16,35, 15,17,37,40,19,18	38	39	"	"	"	"
W30	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,37,16,35, 15,17,37,40,19,38	18	39	"	"	"	"

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TABLE XXIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
W31	1E Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	2	21	52	520	ns	15,16
W32	1E "	2,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	23	21	"	"	"	"
W33	1E "	2,23,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	3	21	"	"	"	"
W34	1E "	2,23,3,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	22	21	"	"	"	"
W35	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	25	4	"	"	"	"
W36	2E "	2,23,3,22,25,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	6	4	"	"	"	"
W37	2E "	2,23,3,22,25,6,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	5	4	"	"	"	"
W38	2E "	2,23,3,22,25,6,5,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	24	4	"	"	"	"
W39	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	7	26	"	"	"	"
W40	3E "	2,23,3,22,25,6,5,24,7,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	28	26	"	"	"	"

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TABLE XXIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W41	3E Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	2,23,3,22,25,6,5,24,7,28,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	8	26	52	520	ns	15,16
W42	3E "	2,23,3,22,25,6,5,24,7,28,8,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	27	26	"	"	"	"
W43	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	11	9	"	"	"	"
W44	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	31	9	"	"	"	"
W45	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	30	9	"	"	"	"
W46	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	29	9	"	"	"	"
W47	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,13,14,33,36,16,35,15, 17,37,40,19,38,18	32	12	"	"	"	"
W48	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,14,33,36,16,35,15, 17,37,40,19,38,18	13	12	"	"	"	"
W49	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,33,36,16,35,15, 17,37,40,19,38,18	14	12	"	"	"	"
W50	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,36,16,35,15, 17,37,40,19,38,18	33	12	"	"	"	"

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TABLE XXIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W51	6E Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,16,35,15, 17,37,40,19,38,18	36	34	52	520	ns	15,16
W52	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,35,15, 17,37,40,19,38,18	16	34	"	"	"	"
W53	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,15, 17,37,40,19,38,18	35	34	"	"	"	"
W54	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 17,37,40,19,38,18	15	34	"	"	"	"
W55	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,37,40,19,38,18	17	20	"	"	"	"
W56	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,40,19,38,18	37	20	"	"	"	"
W57	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,19,38,18	40	20	"	"	"	"
W58	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,38,18	19	39	"	"	"	"
W59	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,18	38	39	"	"	"	"
W60	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38	18	39	"	"	"	"

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TABLE XXIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X1	1E Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	2	21	13	130	ns	15,16
X2	1E "	2,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	23	21	"	"	"	"
X3	1E "	2,23,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	3	21	"	"	"	"
X4	1E "	2,23,3,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	22	21	"	"	"	"
X5	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	25	4	"	"	"	"
X6	2E "	2,23,3,22,25,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	6	4	"	"	"	"
X7	2E "	2,23,3,22,25,6,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	5	4	"	"	"	"
X8	2E "	2,23,3,22,25,6,5,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	24	4	"	"	"	"
X9	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	7	26	"	"	"	"
X10	3E "	2,23,3,22,25,6,5,24,7,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	28	26	"	"	"	"

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TABLE XXIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X11	3E Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	2,23,3,22,25,6,5,24,7,28,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	8	26	13	130	ns	15,16
X12	3E "	2,23,3,22,25,6,5,24,7,28,8,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	27	26	"	"	"	"
X13	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	11	9	"	"	"	"
X14	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	31	9	"	"	"	"
X15	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	30	9	"	"	"	"
X16	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	29	9	"	"	"	"
X17	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,13,14,33,36,16,35,15, 17,37,40,19,38,18	32	12	"	"	"	"
X18	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,14,33,36,16,35,15, 17,37,40,19,38,18	13	12	"	"	"	"
X19	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,33,36,16,35,15, 17,37,40,19,38,18	14	12	"	"	"	"
X20	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,36,16,35,15, 17,37,40,19,38,18	33	12	"	"	"	"

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TABLE XXIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X21	6E Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,16,35,15, 17,37,40,19,38,18	36	34	13	130	ns	15,16
X22	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,35,15, 17,37,40,19,38,18	16	34	"	"	"	"
X23	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,15, 17,37,40,19,38,18	35	34	"	"	"	"
X24	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 17,37,40,19,38,18	15	34	"	"	"	"
X25	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,37,40,19,38,18	17	20	"	"	"	"
X26	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,40,19,38,18	37	20	"	"	"	"
X27	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,19,38,18	40	20	"	"	"	"
X28	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,38,18	19	39	"	"	"	"
X29	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,18	38	39	"	"	"	"
X30	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38	18	39	"	"	"	"

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TABLE XXV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
Y1	1E Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	2	21	13	130	ns	15,16
Y2	1E "	2,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	23	21	"	"	"	"
Y3	1E "	2,23,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	3	21	"	"	"	"
Y4	1E "	2,23,3,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	22	21	"	"	"	"
Y5	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	25	4	"	"	"	"
Y6	2E "	2,23,3,22,25,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	6	4	"	"	"	"
Y7	2E "	2,23,3,22,25,6,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	5	4	"	"	"	"
Y8	2E "	2,23,3,22,25,6,5,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	24	4	"	"	"	"
Y9	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	7	26	"	"	"	"
Y10	3E "	2,23,3,22,25,6,5,24,7,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	28	26	"	"	"	"

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TABLE XXV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
Y11	3E Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	2,23,3,22,25,6,5,24,7,28,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	8	26	13	130	ns	15,16
Y12	3E "	2,23,3,22,25,6,5,24,7,28,8,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	27	26	"	"	"	"
Y13	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	11	9	"	"	"	"
Y14	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	31	9	"	"	"	"
Y15	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	30	9	"	"	"	"
Y16	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	29	9	"	"	"	"
Y17	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,13,14,33,36,16,35,15, 17,37,40,19,38,18	32	12	"	"	"	"
Y18	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,14,33,36,16,35,15, 17,37,40,19,38,18	13	12	"	"	"	"
Y19	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,33,36,16,35,15, 17,37,40,19,38,18	14	12	"	"	"	"
Y20	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,36,16,35,15, 17,37,40,19,38,18	33	12	"	"	"	"

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TABLE XXV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y21	6E Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,16,35,15, 17,37,40,19,38,18	36	34	13	130	ns	15,16
Y22	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,35,15, 17,37,40,19,38,18	16	34	"	"	"	"
Y23	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,15, 17,37,40,19,38,18	35	34	"	"	"	"
Y24	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 17,37,40,19,38,18	15	34	"	"	"	"
Y25	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,37,40,19,38,18	17	20	"	"	"	"
Y26	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,40,19,38,18	37	20	"	"	"	"
Y27	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,19,38,18	40	20	"	"	"	"
Y28	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,38,18	19	39	"	"	"	"
Y29	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,18	38	39	"	"	"	"
Y30	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38	18	39	"	"	"	"

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TABLE XXVI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z1	1E Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	2	21	10	100	ns	15,16
Z2	1E "	2,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	23	21	"	"	"	"
Z3	1E "	2,23,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	3	21	"	"	"	"
Z4	1E "	2,23,3,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	22	21	"	"	"	"
Z5	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	25	4	"	"	"	"
Z6	2E "	2,23,3,22,25,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	6	4	"	"	"	"
Z7	2E "	2,23,3,22,25,6,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	5	4	"	"	"	"
Z8	2E "	2,23,3,22,25,6,5,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	24	4	"	"	"	"
Z9	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	7	26	"	"	"	"
Z10	3E "	2,23,3,22,25,6,5,24,7,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	28	26	"	"	"	"

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TABLE XXVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z11	3E Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	2,23,3,22,25,6,5,24,7,28,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	8	26	10	100	ns	15,16
Z12	3E "	2,23,3,22,25,6,5,24,7,28,8,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	27	26	"	"	"	"
Z13	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	11	9	"	"	"	"
Z14	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	31	9	"	"	"	"
Z15	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	30	9	"	"	"	"
Z16	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	29	9	"	"	"	"
Z17	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,13,14,33,36,16,35,15, 17,37,40,19,38,18	32	12	"	"	"	"
Z18	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,14,33,36,16,35,15, 17,37,40,19,38,18	13	12	"	"	"	"
Z19	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,33,36,16,35,15, 17,37,40,19,38,18	14	12	"	"	"	"
Z20	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,36,16,35,15, 17,37,40,19,38,18	33	12	"	"	"	"

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TABLE XXVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
Z21	6E Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,16,35,15, 17,37,40,19,38,18	36	34	10	100	ns	15,16
Z22	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,35,15, 17,37,40,19,38,18	16	34	"	"	"	"
Z23	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,15, 17,37,40,19,38,18	35	34	"	"	"	"
Z24	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 17,37,40,19,38,18	15	34	"	"	"	"
Z25	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,37,40,19,38,18	17	20	"	"	"	"
Z26	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,40,19,38,18	37	20	"	"	"	"
Z27	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,19,38,18	40	20	"	"	"	"
Z28	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,38,18	19	39	"	"	"	"
Z29	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,18	38	39	"	"	"	"
Z30	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38	18	39	"	"	"	"

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TABLE XXVII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
AA1	1E Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	23,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	2	21	10	100	ns	15,16
AA2	1E "	2,3,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	23	21	"	"	"	"
AA3	1E "	2,23,22,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	3	21	"	"	"	"
AA4	1E "	2,23,3,25,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	22	21	"	"	"	"
AA5	2E "	2,23,3,22,6,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	25	4	"	"	"	"
AA6	2E "	2,23,3,22,25,5,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	6	4	"	"	"	"
AA7	2E "	2,23,3,22,25,6,24,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	5	4	"	"	"	"
AA8	2E "	2,23,3,22,25,6,5,7,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	24	4	"	"	"	"
AA9	3E "	2,23,3,22,25,6,5,24,28,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	7	26	"	"	"	"
AA10	3E "	2,23,3,22,25,6,5,24,7,8,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	28	26	"	"	"	"

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TABLE XXVII. Propagation delay time (t_{PHL}) test - Continued.

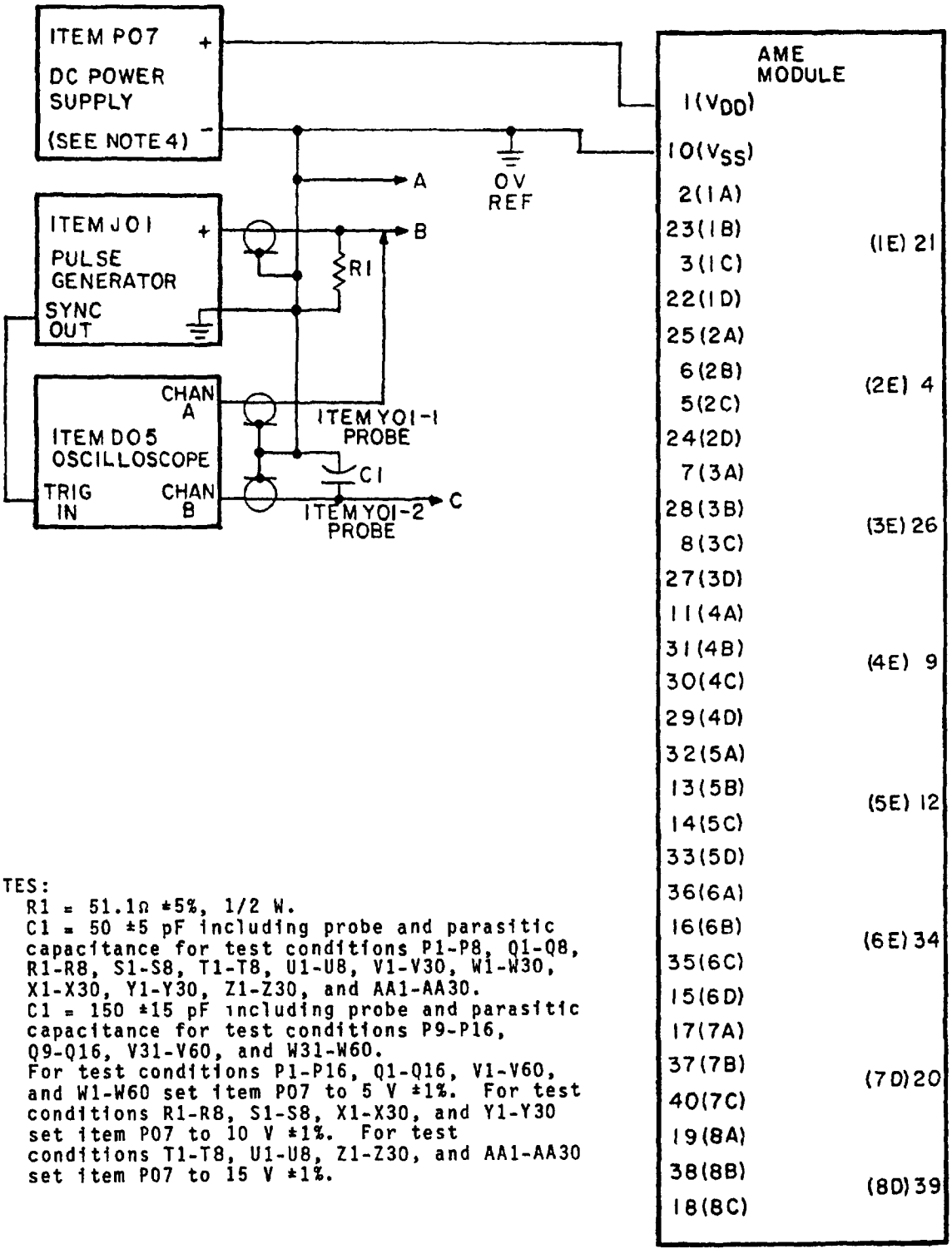
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
AA11	3E Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	2,23,3,22,25,6,5,24,7,28,27,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	8	26	10	100	ns	15,16
AA12	3E "	2,23,3,22,25,6,5,24,7,28,8,11,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	27	26	"	"	"	"
AA13	4E "	2,23,3,22,25,6,5,24,7,28,8,27,31, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	11	9	"	"	"	"
AA14	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 30,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	31	9	"	"	"	"
AA15	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,29,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	30	9	"	"	"	"
AA16	4E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,32,13,14,33,36,16,35,15, 17,37,40,19,38,18	29	9	"	"	"	"
AA17	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,13,14,33,36,16,35,15, 17,37,40,19,38,18	32	12	"	"	"	"
AA18	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,14,33,36,16,35,15, 17,37,40,19,38,18	13	12	"	"	"	"
AA19	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,33,36,16,35,15, 17,37,40,19,38,18	14	12	"	"	"	"
AA20	5E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,36,16,35,15, 17,37,40,19,38,18	33	12	"	"	"	"

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TABLE XXVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°C ±5°C ambient: Min	Initial limits 25°C ±5°C ambient: Max	Unit	Test figure
AA21	6E Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,16,35,15, 17,37,40,19,38,18	36	34	10	100	ns	15,16
AA22	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,35,15, 17,37,40,19,38,18	16	34	"	"	"	"
AA23	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,15, 17,37,40,19,38,18	35	34	"	"	"	"
AA24	6E "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 17,37,40,19,38,18	15	34	"	"	"	"
AA25	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,37,40,19,38,18	17	20	"	"	"	"
AA26	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,40,19,38,18	37	20	"	"	"	"
AA27	7D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,19,38,18	40	20	"	"	"	"
AA28	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,38,18	19	39	"	"	"	"
AA29	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,18	38	39	"	"	"	"
AA30	8D "	2,23,3,22,25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36,16,35, 15,17,37,40,19,38	18	39	"	"	"	"

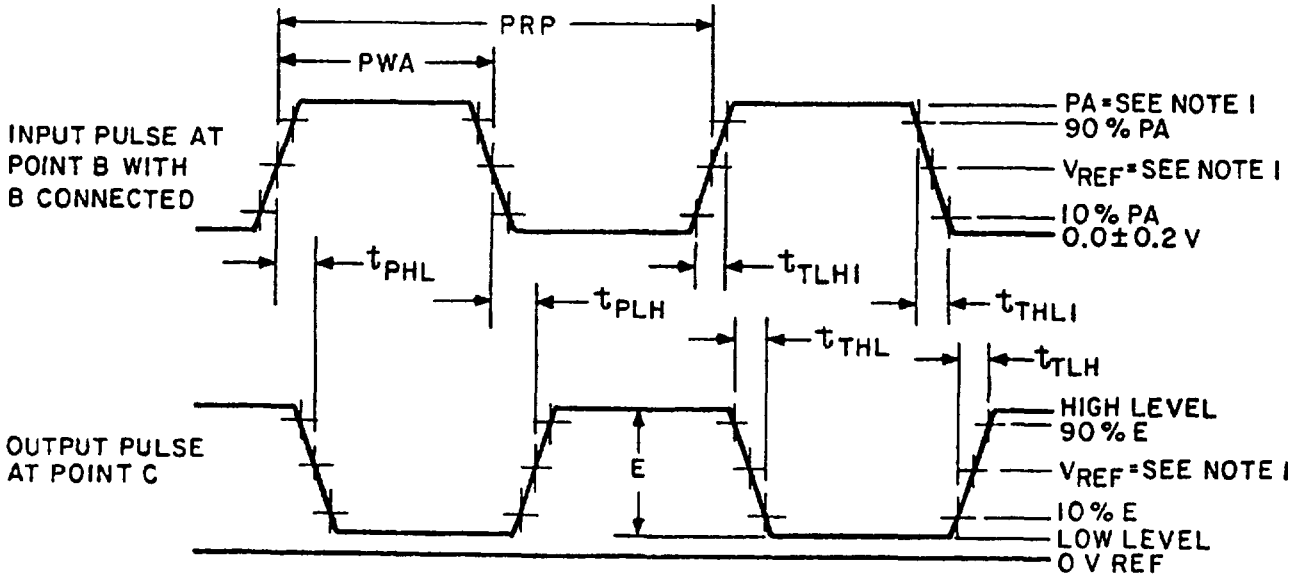
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. C1 = 50 ±5 pF including probe and parasitic capacitance for test conditions P1-P8, Q1-Q8, R1-R8, S1-S8, T1-T8, U1-U8, V1-V30, W1-W30, X1-X30, Y1-Y30, Z1-Z30, and AA1-AA30.
3. C1 = 150 ±15 pF including probe and parasitic capacitance for test conditions P9-P16, Q9-Q16, V31-V60, and W31-W60.
4. For test conditions P1-P16, Q1-Q16, V1-V60, and W1-W60 set item P07 to 5 V ±1%. For test conditions R1-R8, S1-S8, X1-X30, and Y1-Y30 set item P07 to 10 V ±1%. For test conditions T1-T8, U1-U8, Z1-Z30, and AA1-AA30 set item P07 to 15 V ±1%.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVI through XXVII.

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NOTES:

1. Input pulse parameters:
   PA = 5.0 ±0.1 V for test conditions P1-P16, Q1-Q16, V1-V60, and W1-W60.
   = 10.0 ±0.1 V for test conditions R1-R8, S1-S8, X1-X30, and Y1-Y30.
   = 15.0 ±0.1 V for test conditions T1-T8, U1-U8, Z1-Z30, and AA1-AA30.
   V_{REF} = 2.5 ±0.1 V for test conditions P1-P16, Q1-Q16, V1-V60, and W1-W60.
   = 5.0 ±0.1 V for test conditions R1-R8, S1-S8, X1-X30, and Y1-Y30.
   = 7.5 ±0.1 V for test conditions T1-T8, U1-U8, Z1-Z30, and AA1-AA30.
   PWA = 2.5 ±0.2 µs for test conditions P1-P16, Q1-Q16, V1-V60, and W1-W60.
   = 1.0 ±0.1 µs for all other test conditions.
   PRP = 5.0 ±0.5 µs.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Before making the measurements verify that the output pulse high level is $\geq$ 4.8 V for test conditions P1-P16, Q1-Q16, V1-V60, and W1-W60; $\geq$ 9.5 V for test conditions R1-R8, S1-S8, X1-X30, and Y1-Y30 and $\geq$ 13.5 V for test conditions T1-T8, U1-U8, Z1-Z30, and AA1-AA30 and the low level is $\leq$ 0.4 V for test conditions P1-P16, Q1-Q16, V1-V60, and W1-W60; $\leq$ 0.5 V for test conditions R1-R8, S1-S8, X1-X30, and Y1-Y30 and $\leq$ 1.5 V for test conditions T1-T8, U1-U8, Z1-Z30, and AA1-AA30.
3. The lead length of electrical conductors between resistor R1 and input pins, items Y01-1 and Y01-2 to input and output pins, and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI through XXVII.

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TABLE XXVIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB	Power supply filter capacitance	Open	1	0.007	0.013	µF	17

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TABLE XXIX. Input capacitance test.

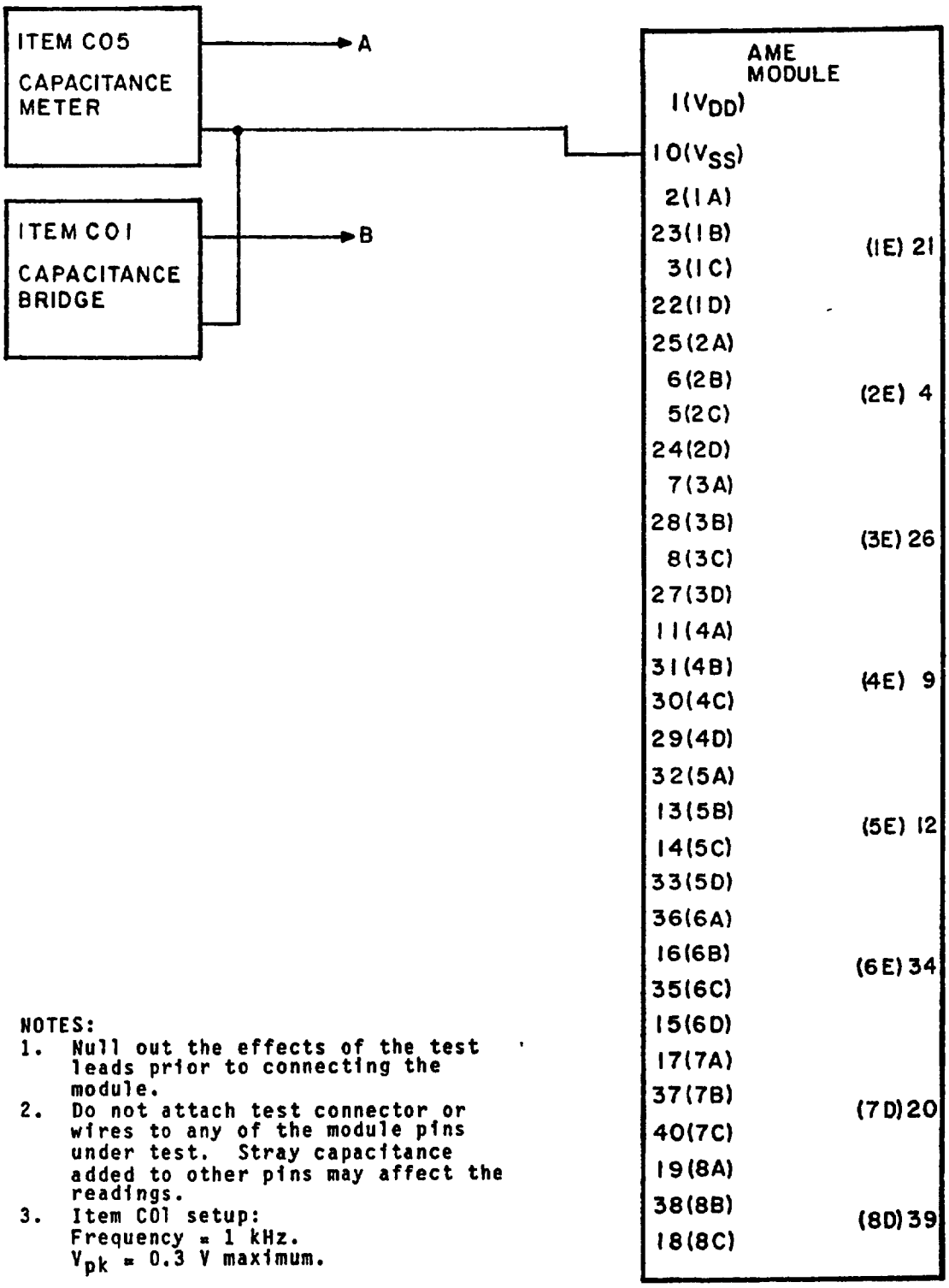
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC1	1A Input capacitance	2	Open	2	15	pF	17
CC2	1B "	23	"	"	"	"	"
CC3	1C "	3	"	"	"	"	"
CC4	1D "	22	"	"	"	"	"
CC5	2A "	25	"	"	"	"	"
CC6	2B "	6	"	"	"	"	"
CC7	2C "	5	"	"	"	"	"
CC8	2D "	24	"	"	"	"	"
CC9	3A "	7	"	"	"	"	"
CC10	3B "	28	"	"	"	"	"
CC11	3C "	8	"	"	"	"	"
CC12	3D "	27	"	"	"	"	"
CC13	4A "	11	"	"	"	"	"
CC14	4B "	31	"	"	"	"	"
CC15	4C "	30	"	"	"	"	"
CC16	4D "	29	"	"	"	"	"
CC17	5A "	32	"	"	"	"	"
CC18	5B "	13	"	"	"	"	"
CC19	5C "	14	"	"	"	"	"
CC20	5D "	33	"	"	"	"	"

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TABLE XXIX. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC21	6A Input capacitance	36	Open	2	15	pF	17
CC22	6B "	16	"	"	"	"	"
CC23	6C "	35	"	"	"	"	"
CC24	6D "	15	"	"	"	"	"
CC25	7A "	17	"	"	"	"	"
CC26	7B "	37	"	"	"	"	"
CC27	7C "	40	"	"	"	"	"
CC28	8A "	19	"	"	"	"	"
CC29	8B "	38	"	"	"	"	"
CC30	8C "	18	"	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

* FIGURE 17. Power supply filter and input capacitance test circuit for tables XXVIII and XXIX.

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